MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, $J\overline{K}$, DIGITAL, KEY CODE GBB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic $J\overline{K}$ flip-flop module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.925 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GBB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/143-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

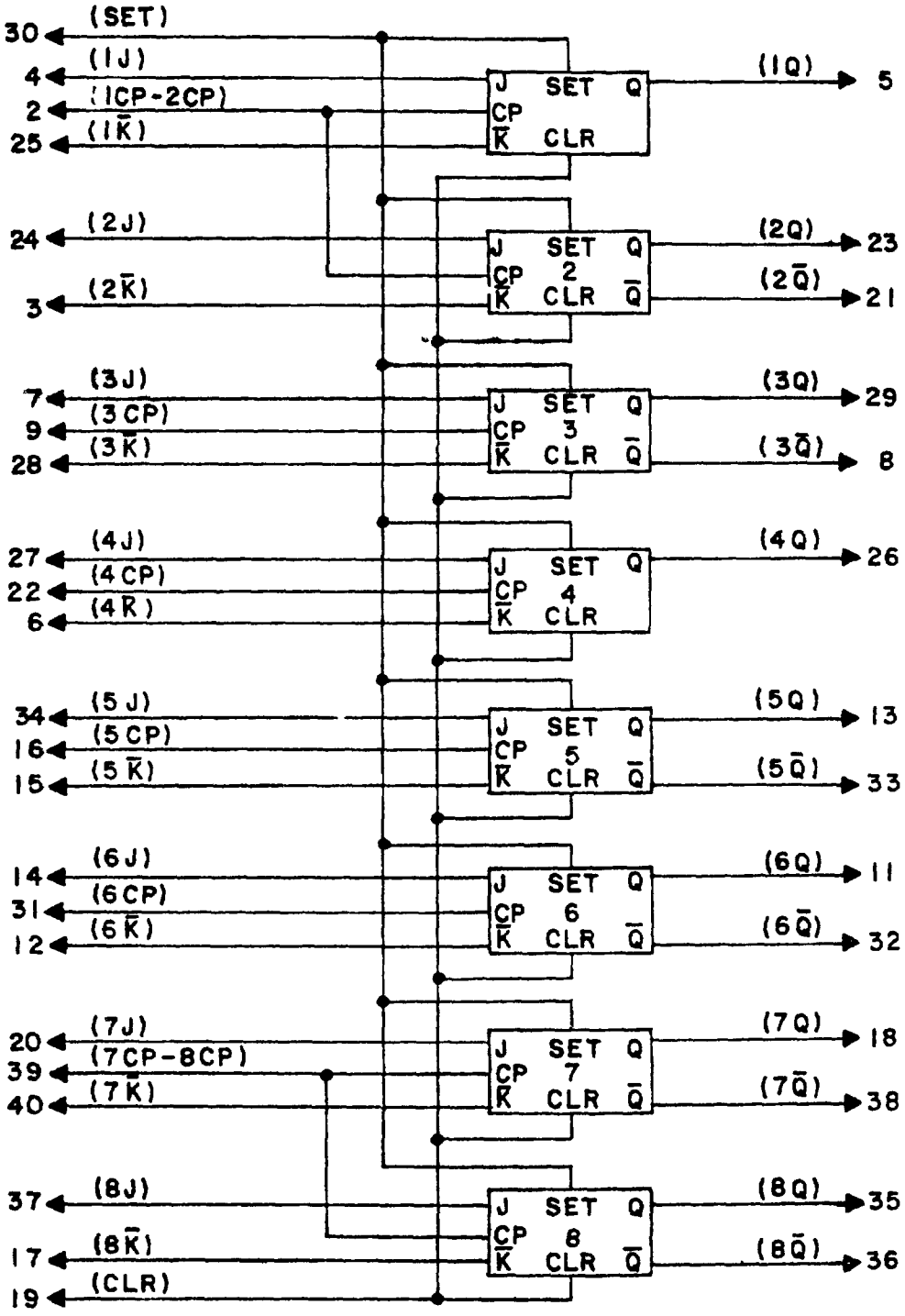
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. 1J-8J, 1$\overline{K}$-8$\overline{K}$ = Data inputs.
2. 1CP-2CP, 3CP-6CP, 7CP-8CP = Clock inputs.
3. SET = Common set input.
4. CLR = Common clear input.
5. 1Q-8Q = Data outputs.
6. 2$\overline{Q}$, 3$\overline{Q}$, 5$\overline{Q}$-8$\overline{Q}$ = Data outputs.

* FIGURE 1. GBB module function diagram, J$\overline{K}$ flip-flop.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B14	V_{OUT} (low loaded)	IV	S	10	800	10	800	10	800	10	800	mV
C1,C2, C24-C26	Input breakdown current	V	S	0	2	0	2	0	2	0	2	mA
C3-C22, C27-C30	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
C23	Input breakdown current	V	S	0	4	0	4	0	4	0	4	mA
D1-D14	Output short circuit current	VI	X	-30	-85	-30	-85	-30	-85	-30	-85	mA
E1-E24	Input clamp diode voltage	VII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
F1-F103	Functional (input threshold check)	VIII	X	← Test vectors must produce output vectors as specified in test table VIII. →								
G1-G14	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	mV
H1-H14	Output voltage, high level (input threshold check)	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J3, J25,J26	Input current, high level	X	X	0	160	0	160	0	160	0	160	µA
J4-J8, J27-J30	Input current, high level	X	X	0	80	0	80	0	80	0	80	µA
J9-J24	Input current, high level	X	X	0	40	0	40	0	40	0	40	µA
K1-K3	Input current, low level	XI	X	-0.40	-6.92	-0.40	-6.92	-0.40	-6.92	-0.40	-6.92	mA
K4-K8	Input current, low level	XI	X	-0.20	-3.46	-0.20	-3.46	-0.20	-3.46	-0.20	-3.46	mA
K9-K24	Input current, low level	XI	X	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	-0.10	-1.73	mA
L	Power supply current	XII	S	30	144	30	144	30	144	30	144	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M1-M14	Transition time (t_{TLH}) (setup and hold time check), C_L = 50 pF	XIII	S	4	30	4	30	4	35	4	35	ns
M15-M28	Transition time (t_{TLH}) (setup and hold time check), C_L = 150 pF	XIII	Q	4	50	4	50	4	55	4	55	ns
N1-N14	Transition time (t_{THL}) (setup and hold time check), C_L = 50 pF	XIII	S	4	13	4	13	4	18	4	18	ns
N15-N28	Transition time (t_{THL}) (setup and hold time check), C_L = 150 pF	XIII	Q	4	20	4	20	4	25	4	25	ns
P1-P14	Propagation delay time from clock (t_{PLH}), C_L = 50 pF	XIV	S	6	30	6	30	6	32	6	32	ns
P15-P28	Propagation delay time from clock (t_{PLH}), C_L = 150 pF	XIV	Q	9	37	9	37	9	40	9	40	ns
Q1-Q14	Propagation delay time from clock (t_{PHL}), C_L = 50 pF	XIV	S	8	40	8	40	8	42	8	42	ns
Q15-Q28	Propagation delay time from clock (t_{PHL}), C_L = 150 pF	XIV	Q	10	58	10	58	10	60	10	60	ns
R1-R8	Propagation delay time from set (t_{PLH}), C_L = 50 pF	XV	S	8	35	8	35	8	38	8	38	ns
R9-R16	Propagation delay time from set (t_{PLH}), C_L = 150 pF	XV	Q	8	42	8	42	8	45	8	45	ns
S1-S6	Propagation delay time from set (t_{PHL}), C_L = 50 pF	XV	S	15	64	15	64	15	67	15	67	ns
S7-S12	Propagation delay time from set (t_{PHL}), C_L = 150 pF	XV	Q	18	75	18	75	18	79	18	79	ns

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See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
T1-T6	Propagation delay time from clear (t_{PLH}), C_L = 50 pF	XV	S	8	35	8	35	8	38	8	38	ns
T7-T12	Propagation delay time from clear (t_{PLH}), C_L = 150 pF	XV	Q	8	42	8	42	8	45	8	45	ns
U1-U8	Propagation delay time from clear (t_{PHL}), C_L = 50 pF	XV	S	15	60	15	60	15	63	15	63	ns
U9-U16	Propagation delay time from clear (t_{PHL}), C_L = 150 pF	XV	Q	17	68	17	68	17	71	17	71	ns
V1-V14	Maximum operating frequency check (measuring output pulse period-t_{PW}), C_L = 50 pF	XVI	S	110	130	110	130	110	130	110	130	ns
W1-W14	Output crosstalk, low level	XVII	Q	0	400	0	400	0	400	0	400	mV
X1-X14	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
Y	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	mV
Z	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
AA1	Input capacitance	XX	Q	5	45	5	45	5	45	5	45	pF
AA2-AA4	Input capacitance	XX	Q	5	30	5	30	5	30	5	30	pF
AA5-AA24	Input capacitance	XX	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through Z shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through AA shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military specification MIL-M-28787/143 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833758. Part number M28787/143-1 supersedes part numbers 2833758 and M28787/143.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table 1.

Supply voltage	5.0 V dc ±10 percent
Maximum power dissipation	792 mW maximum
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High Level	2 V minimum, V_{CC} maximum

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Input current, CLR		
Low level	(V_{IL} = 0.0 V)	-6.92 mA maximum
	(V_{IL} = 0.4 V)	-6.4 mA maximum
High Level	(V_{IH} = 2.4 V)	160 µA maximum
	(V_{IH} = 5.5 V)	4 mA maximum
Input current, CP1-CP2, CP7-CP8		
Low level	(V_{IL} = 0.0 V)	-6.92 mA maximum
	(V_{IL} = 0.4 V)	-6.4 mA maximum
High Level	(V_{IH} = 2.4 V)	160 µA maximum
	(V_{IH} = 5.5 V)	2 mA maximum
Input current, SET		
Low level	(V_{IL} = 0.0 V)	-3.46 mA maximum
	(V_{IL} = 0.4 V)	-3.2 mA maximum
High Level	(V_{IH} = 2.4 V)	80 µA maximum
	(V_{IH} = 5.5 V)	2 mA maximum
Input current, CP3-CP6		
Low level	(V_{IL} = 0.0 V)	-3.46 mA maximum
	(V_{IL} = 0.4 V)	-3.2 mA maximum
High Level	(V_{IH} = 2.4 V)	80 µA maximum
	(V_{IH} = 5.5 V)	1 mA maximum
Input current, J, $\overline{K}$		
Low level	(V_{IL} = 0.0 V)	-1.73 mA maximum
	(V_{IL} = 0.4 V)	-1.6 mA maximum
High Level	(V_{IH} = 2.4 V)	40 µA maximum
	(V_{IH} = 5.5 V)	1 mA maximum
Output voltage		
Low level	(I_{OL} = 16 mA)	0.4 V maximum
High Level	(I_{OH} = -800 µA)	2.4 V minimum
Output drive current maximum capability		
Low level	(V_{OL} = 0.4 V)	16 mA
High Level	(V_{OH} = 2.4 V)	-800 µA
Propagation delay time (t_{PLH})		
CP to Q, $\overline{Q}$		
	50 pF load	6 ns minimum, 32 ns maximum
	150 pF load	9 ns minimum, 40 ns maximum
SET to Q		
	50 pF load	8 ns minimum, 38 ns maximum
	150 pF load	8 ns minimum, 45 ns maximum
CLR to $\overline{Q}$		
	50 pF load	8 ns minimum, 38 ns maximum
	150 pF load	8 ns minimum, 45 ns maximum
Propagation delay time (t_{PHL})		
CP to Q, $\overline{Q}$		
	50 pF load	8 ns minimum, 42 ns maximum
	150 pF load	10 ns minimum, 60 ns maximum
SET to $\overline{Q}$		
	50 pF load	15 ns minimum, 67 ns maximum
	150 pF load	18 ns minimum, 79 ns maximum
CLR to Q		
	50 pF load	15 ns minimum, 63 ns maximum
	150 pF load	17 ns minimum, 71 ns maximum
Maximum operating frequency		
f_{MAX} (50 pF load)		16.6 MHz maximum
Setup time		
t_{setup} (J, $\overline{K}$)		20 ns minimum

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Hold time
t_{hold} (J, $\overline{K}$) 6 ns minimum

Pulse width
t_{pW} (CP) 20 ns minimum
t_{pW} (SET, CLR) 20 ns minimum

* 6.5 Module function. The J$\overline{K}$ flip-flop module contains eight J$\overline{K}$ flip-flops with common buffered asynchronous SET and CLEAR inputs (active high) for all flip-flops. The clock input is common to flip-flops 1 and 2 (CP1,CP2) and to flip-flops 7 and 8 (CP7-CP8). Flip-flops 3-6 each have an independent clock input. Flip-flops 2, 3, 5, 6, 7, and 8 have both a Q and $\overline{Q}$ output. Flip-Flops 1 and 4 have only a Q output. Information on the inputs is transferred to the output on the positive clock input transition.

6.5.1 Input to output relationship. Each J$\overline{K}$ FLIP-FLOP has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/-5/

Inputs					Outputs	
SET	CLEAR	CLOCK	J	$\overline{K}$	Q	$\overline{Q}$
H	L	X	X	X	H	L
L	H	X	X	X	L	H
H	H	X	X	X	H	H
					6/	6/
L	L		L	L	$\overline{L}$	$\overline{H}$
L	L		H	L	Toggle	
L	L		L	H	Q_o	$\overline{Q}_o$
L	L		H	H	H	$\overline{L}$
L	L	L	X	X	Q_o	$\overline{Q}_o$

1/ H = High-level (steady state).

2/ L = Low-level (steady state).

3/ X = Irrelevant (any input, including transitions).

4/ = Transition from low to high.

5/ Q_o, $\overline{Q}_o$ = The level of Q, $\overline{Q}$ respectively, before the indicated steady-state input conditions were established.

6/ This configuration is unstable; that is, it will not persist when SET and CLEAR return to the inactive (low-level) state.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

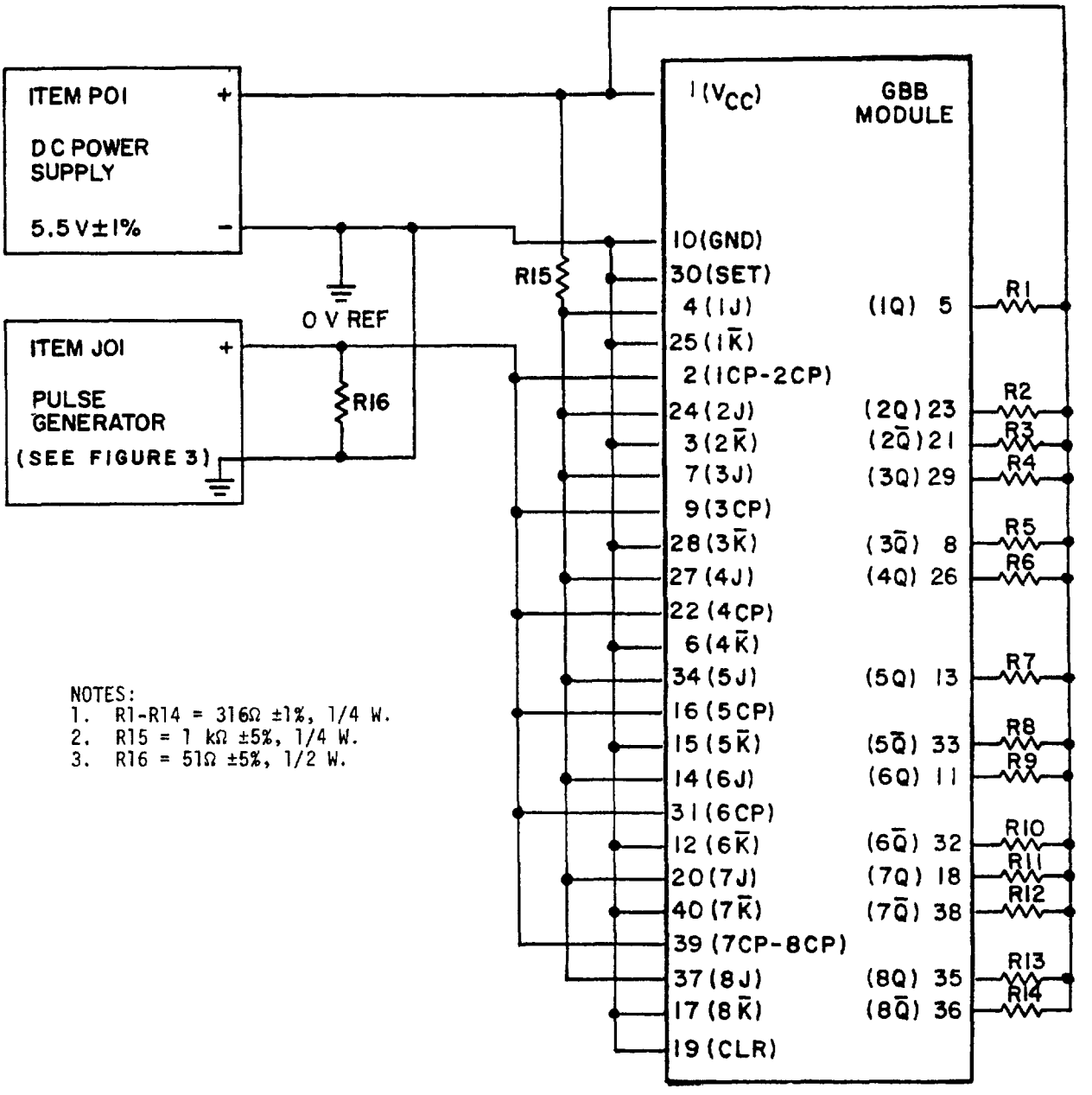
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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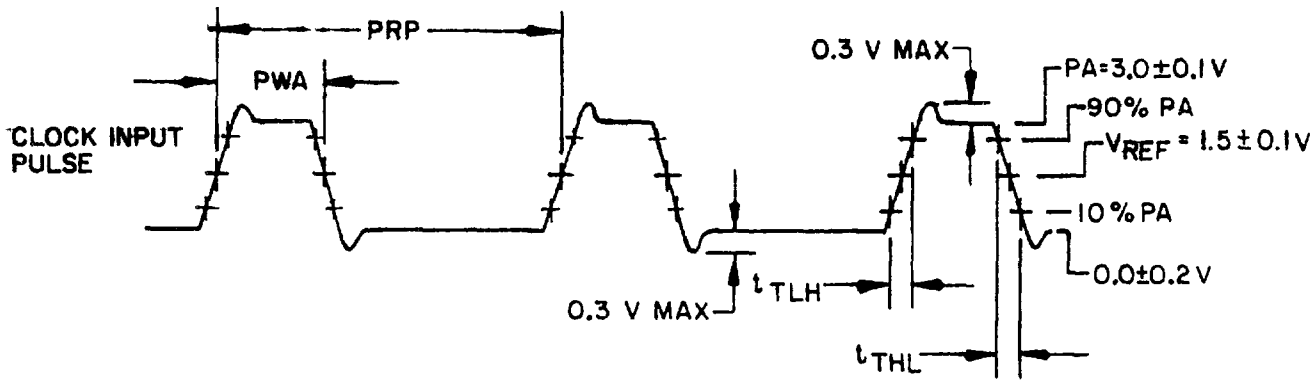
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NOTES:
1. R1-R14 = 316Ω ±1%, 1/4 W.
2. R15 = 1 kΩ ±5%, 1/4 W.
3. R16 = 51Ω ±5%, 1/2 W.

* FIGURE 2. Operating temperature and life test circuit

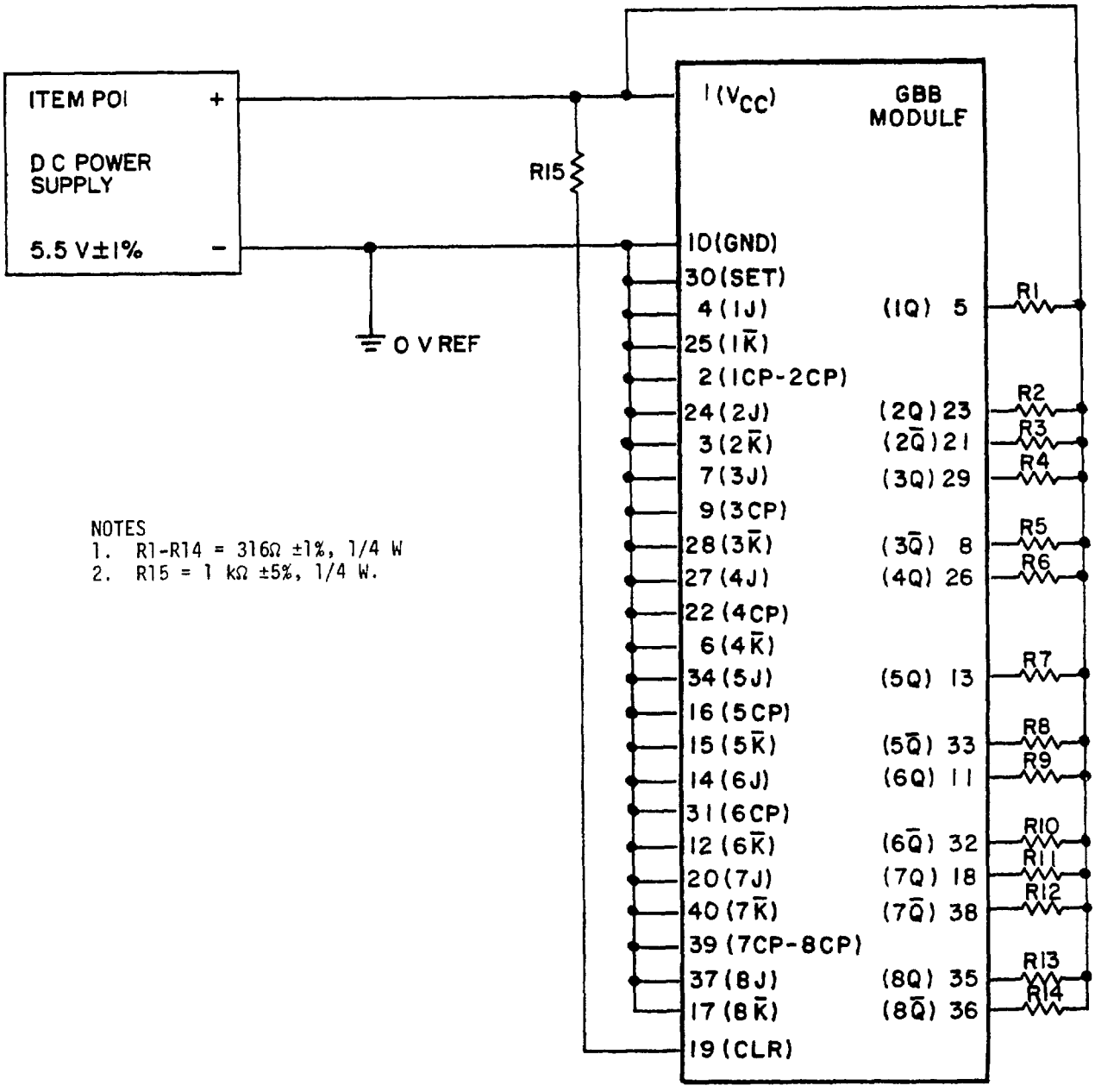
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NOTES:
1. Clock input pulse parameters:
   PRP = 1,000 ±200 ns
   PWA = 250 ±50 ns
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltages are referenced to ground.

FIGURE 3 Operating temperature and life test circuit waveforms.

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NOTES
1. R1-R14 = 316Ω ±1%, 1/4 W
2. R15 = 1 kΩ ±5%, 1/4 W.

FIGURE 4. Component temperature test circuit.

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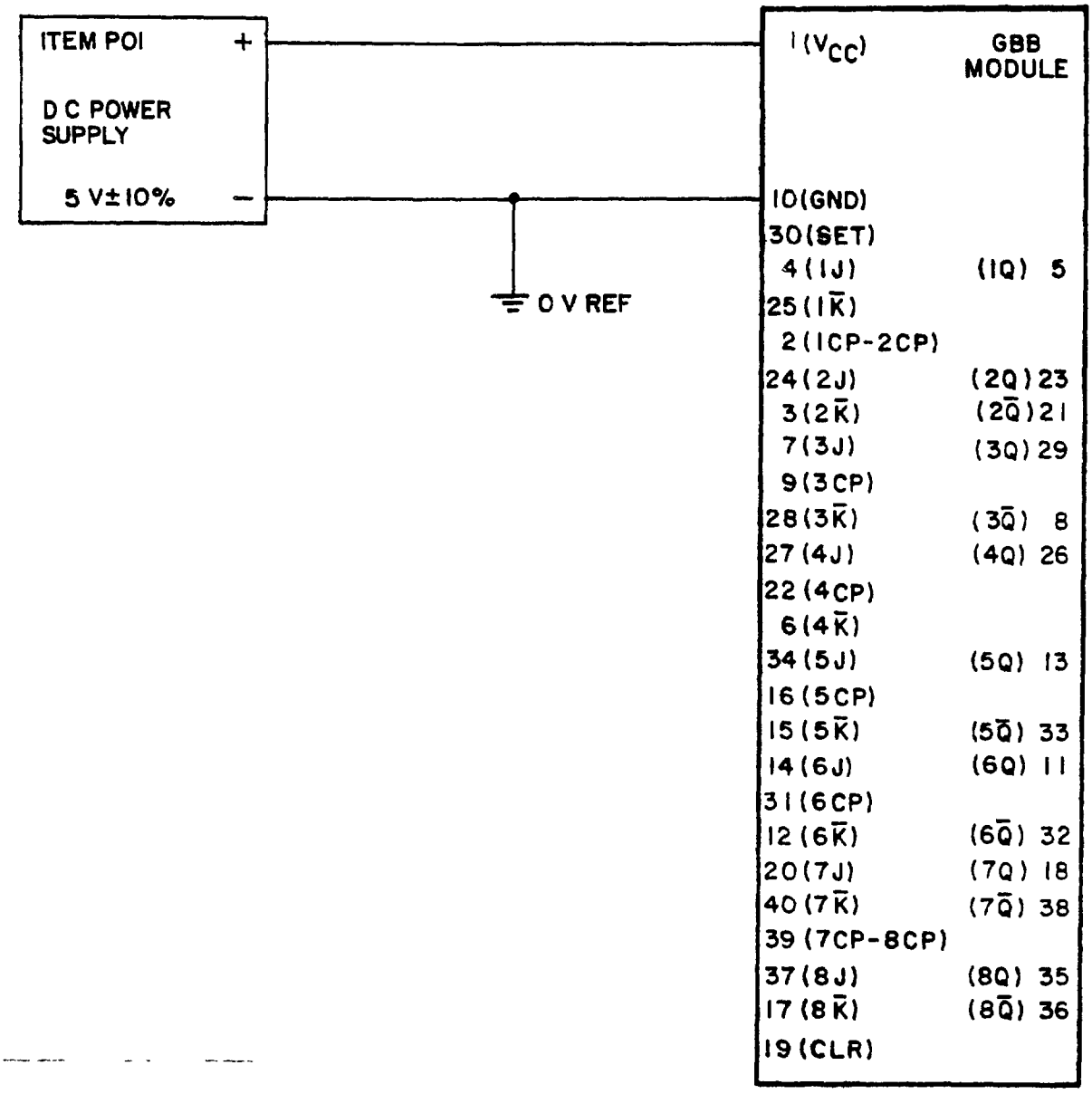


FIGURE 5. Vibration test circuit.

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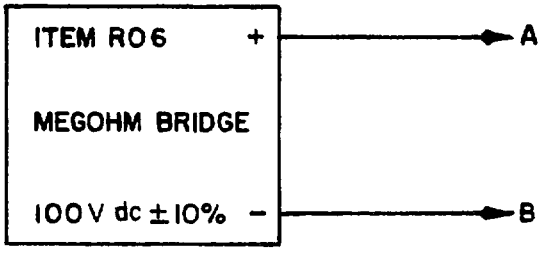
TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	mΩ	6

* TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
B1	1Q	V_{OUT} (low loaded)	5	19	30	10	800	mV	7
B2	2Q	"	23	"	"	"	"	"	"
B3	3Q	"	29	"	"	"	"	"	"
B4	4Q	"	26	"	"	"	"	"	"
B5	5Q	"	13	"	"	"	"	"	"
B6	6Q	"	11	"	"	"	"	"	"
B7	7Q	"	18	"	"	"	"	"	"
B8	8Q	"	35	"	"	"	"	"	"
B9	$2\overline{Q}$	"	21	30	19	"	"	"	"
B10	$3\overline{Q}$	"	8	"	"	"	"	"	"
B11	$5\overline{Q}$	"	33	"	"	"	"	"	"
B12	$6\overline{Q}$	"	32	"	"	"	"	"	"
B13	$7\overline{Q}$	"	38	"	"	"	"	"	"
B14	$8\overline{Q}$	"	36	"	"	"	"	"	"

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WARNING·

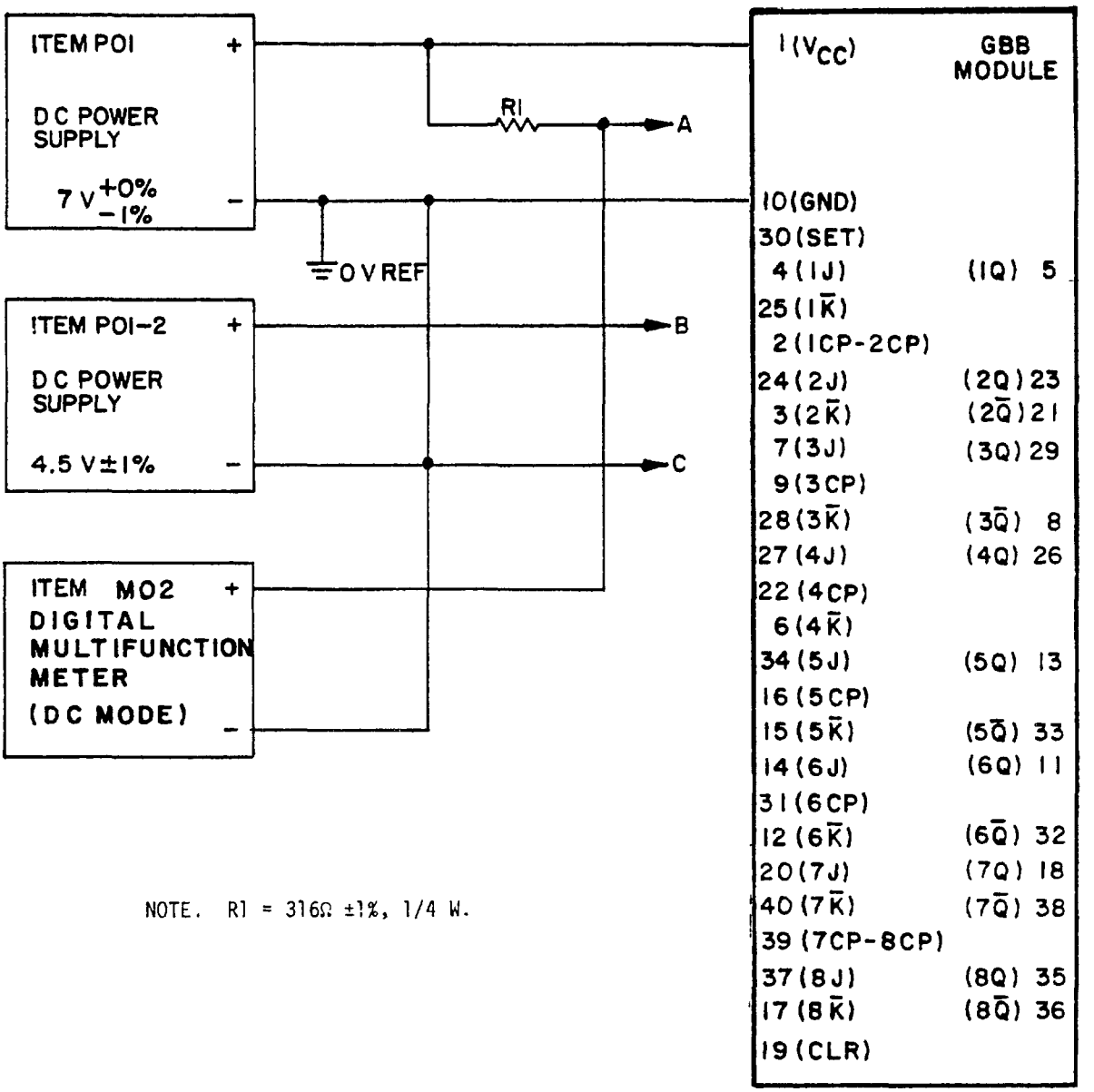
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

1 (V_{CC})	GBB MODULE
10 (GND)	
30 (SET)	
4 (1J)	(1Q) 5
25 (1$\bar{K}$)	
2 (1CP-2CP)	
24 (2J)	(2Q) 23
3 (2$\bar{K}$)	(2$\bar{Q}$) 21
7 (3J)	(3Q) 29
9 (3CP)	
28 (3$\bar{K}$)	(3$\bar{Q}$) 8
27 (4J)	(4Q) 26
22 (4CP)	
6 (4$\bar{K}$)	
34 (5J)	(5Q) 13
16 (5CP)	
15 (5$\bar{K}$)	(5$\bar{Q}$) 33
14 (6J)	(6Q) 11
31 (6CP)	
12 (6$\bar{K}$)	(6$\bar{Q}$) 32
20 (7J)	(7Q) 18
40 (7$\bar{K}$)	(7$\bar{Q}$) 38
39 (7CP-8CP)	
37 (8J)	(8Q) 35
17 (8$\bar{K}$)	(8$\bar{Q}$) 36
19 (CLR)	

FIGURE 6. Isolation test circuit for table III.

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NOTE. R1 = 316Ω ±1%, 1/4 W.

* FIGURE 7. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1CP-2CP Input breakdown current	3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	2	0	2	mA	8
C2	7CP-8CP "	2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 31,34,37,40	39	"	2	"	"
C3	3CP "	2,3,4,6,7,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	9	"	1	"	"
C4	4CP "	2,3,4,6,7,9,12,14,15,16, 17,20,24,25,27,28,30,31, 34,37,39,40	22	"	"	"	"
C5	5CP "	2,3,4,6,7,9,12,14,15,17, 20,22,24,25,27,28,30,31, 34,37,39,40	16	"	"	"	"
C6	6CP "	2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 34,37,39,40	31	"	"	"	"
C7	1J "	2,3,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	4	"	"	"	"
C8	2J "	2,3,4,6,7,9,12,14,15,16, 17,20,22,25,27,28,30,31, 34,37,39,40	24	"	"	"	"
C9	3J "	2,3,4,6,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	7	"	"	"	"
C10	4J "	2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,28,30,31, 34,37,39,40	27	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

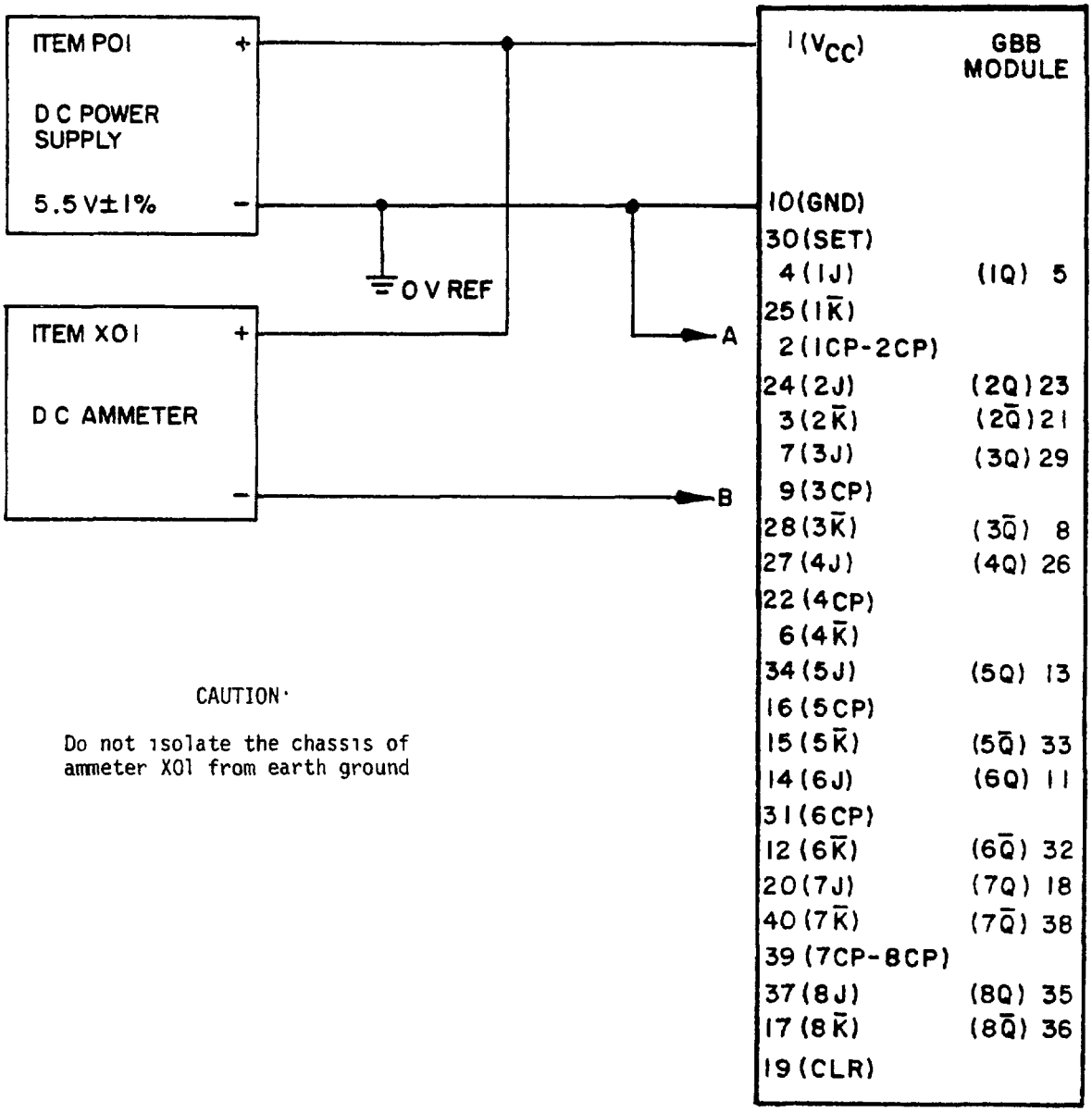
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C11	5J	Input breakdown current	2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 31,37,39,40	34	0	1	mA	8
C12	6J	"	2,3,4,6,7,9,12,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	14	"	"	"	"
C13	7J	"	2,3,4,6,7,9,12,14,15,16, 17,22,24,25,27,28,30,31, 34,37,39,40	20	"	"	"	"
C14	8J	"	2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 31,34,39,40	37	"	"	"	"
C15	1$\overline{K}$	"	2,3,4,6,7,9,12,14,15,16, 17,20,22,24,27,28,30,31, 34,37,39,40	25	"	"	"	"
C16	2$\overline{K}$	"	2,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	3	"	"	"	"
C17	3$\overline{K}$	"	2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,30,31, 34,37,39,40	28	"	"	"	"
C18	4$\overline{K}$	"	2,3,4,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	6	"	"	"	"
C19	5$\overline{K}$	"	2,3,4,6,7,9,12,14,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	15	"	"	"	"
C20	6$\overline{K}$	"	2,3,4,6,7,9,14,15,16,17, 20,22,24,25,27,28,30,31, 34,37,39,40	12	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C21	$7\overline{K}$	Input breakdown current	2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 31,34,37,39	40	0	1	mA	8
C22	$8\overline{K}$	"	2,3,4,6,7,9,12,14,15,16, 20,22,24,25,27,28,30,31, 34,37,39,40	17	"	1	"	"
C23	CLR	"	2,3,4,6,7,9,12,14,15,16, 17,20,22,24,25,27,28,30, 31,34,37,39,40	19	"	4	"	"
C24	SET	"	2,3,4,6,7,9,12,14,15,16, 17,19,20,22,24,25,27,28, 31,34,37,39,40	30	"	2	"	"
C25	1CP-2CP	"	3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,31, 34,37,39,40	2	"	"	"	"
C26	7CP-8CP	"	2,3,4,6,7,9,12,14,15,16, 17,19,20,22,24,25,27,28, 31,34,37,40	39	"	"	"	"
C27	3CP	"	2,3,4,6,7,12,14,15,16,17, 19,20,22,24,25,27,28,31, 34,37,39,40	9	"	1	"	"
C28	4CP	"	2,3,4,6,7,9,12,14,15,16, 17,19,20,24,25,27,28,31, 34,37,39,40	22	"	"	"	"
C29	5CP	"	2,3,4,6,7,9,12,14,15,17, 19,20,22,24,25,27,28,31, 34,37,39,40	16	"	"	"	"
C30	6CP	"	2,3,4,6,7,9,12,14,15,16, 17,19,20,22,24,25,27,28, 34,37,39,40	31	"	"	"	"

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground

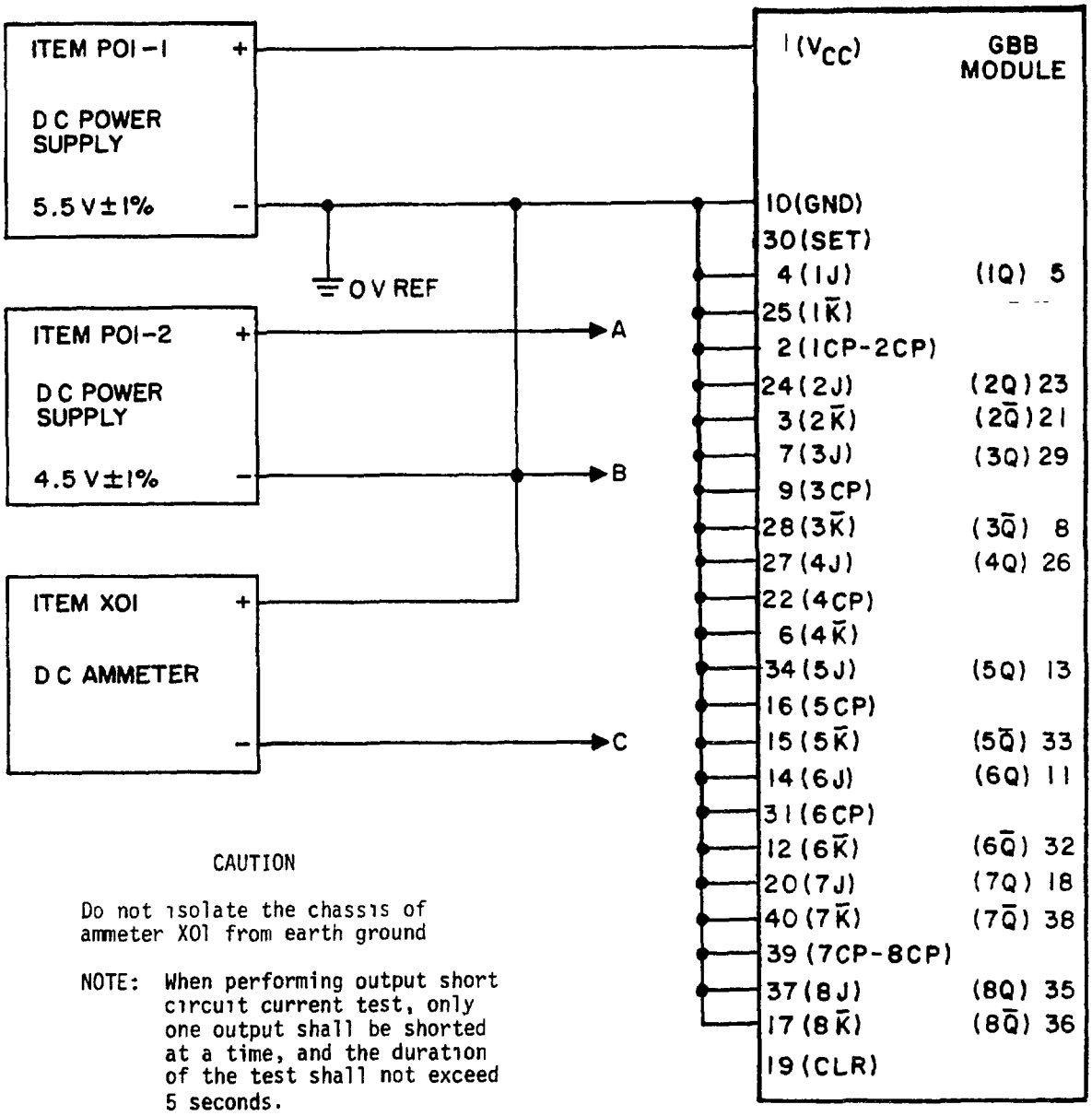
FIGURE 8. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
D1	1Q	Output short circuit current	30	19	5	-30	-85	mA	9
D2	2Q	"	"	"	23	"	"	"	"
D3	3Q	"	"	"	29	"	"	"	"
D4	4Q	"	"	"	26	"	"	"	"
D5	5Q	"	"	"	13	"	"	"	"
D6	6Q	"	"	"	11	"	"	"	"
D7	7Q	"	"	"	18	"	"	"	"
D8	8Q	"	"	"	35	"	"	"	"
D9	$2\overline{Q}$	"	19	30	21	"	"	"	"
D10	$3\overline{Q}$	"	"	"	8	"	"	"	"
D11	$5\overline{Q}$	"	"	"	33	"	"	"	"
D12	$6\overline{Q}$	"	"	"	32	"	"	"	"
D13	$7\overline{Q}$	"	"	"	38	"	"	"	"
D14	$8\overline{Q}$	"	"	"	36	"	"	"	"

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CAUTION

Do not isolate the chassis of ammeter XOI from earth ground

NOTE: When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed 5 seconds.

FIGURE 9. Output short circuit current test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

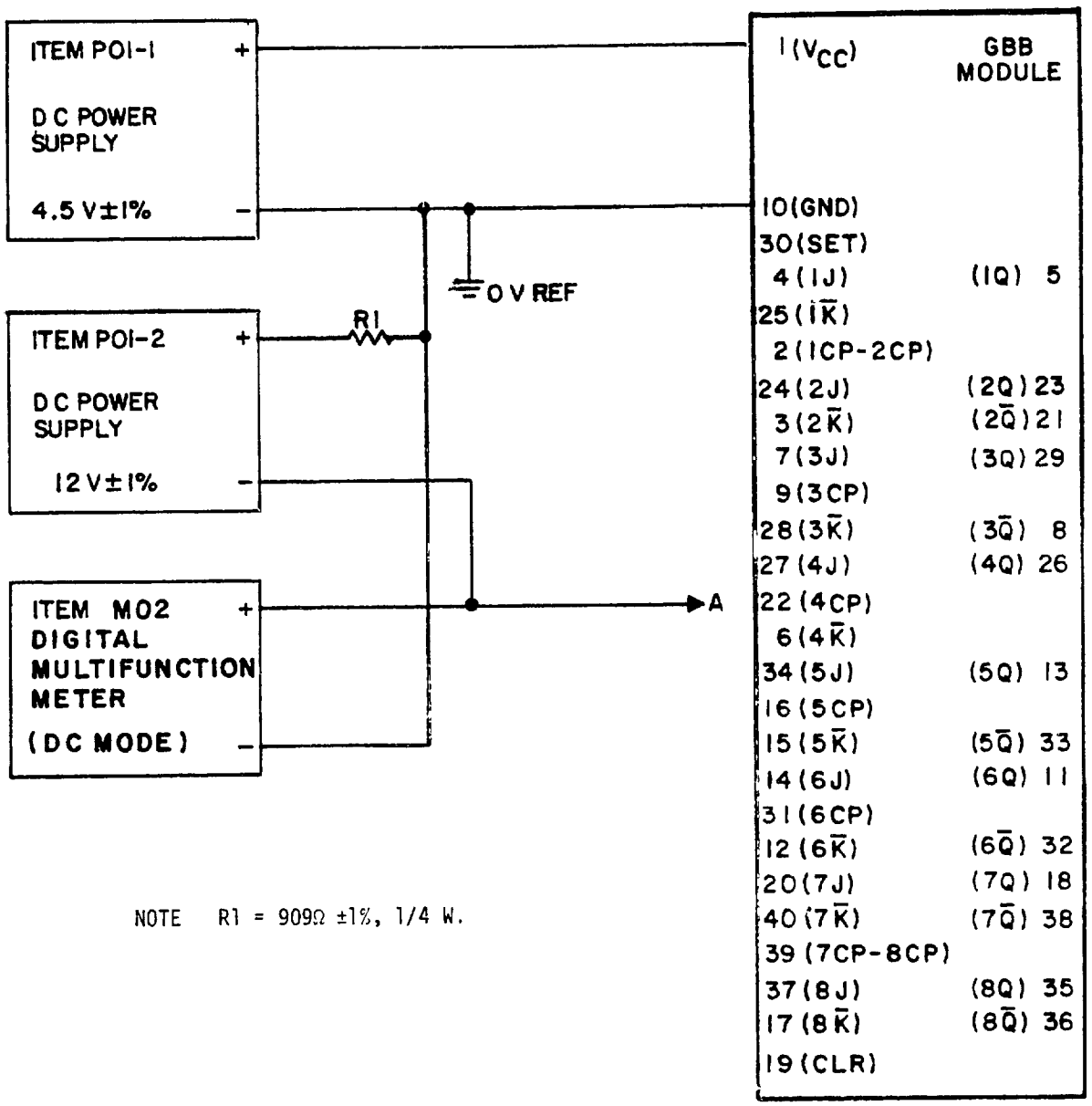
Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
E1	SET	Input clamp diode voltage	30	-0.4	-1.5	V	10
E2	CLR	"	19	"	"	"	"
E3	1CP-2CP	"	2	"	"	"	"
E4	3CP	"	9	"	"	"	"
E5	4CP	"	22	"	"	"	"
E6	5CP	"	16	"	"	"	"
E7	6CP	"	31	"	"	"	"
E8	7CP-8CP	"	39	"	"	"	"
E9	1J	"	4	"	"	"	"
E10	2J	"	24	"	"	"	"
E11	3J	"	7	"	"	"	"
E12	4J	"	27	"	"	"	"
E13	5J	"	34	"	"	"	"
E14	6J	"	14	"	"	"	"
E15	7J	"	20	"	"	"	"
E16	8J	"	37	"	"	"	"
E17	$1\overline{K}$	"	25	"	"	"	"
E18	$2\overline{K}$	"	3	"	"	"	"

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TABLE VII. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
E19	$3\overline{K}$	Input clamp diode voltage	28	-0.4	-1.5	V	10
E20	$4\overline{K}$	"	6	"	"	"	"
E21	$5\overline{K}$	"	15	"	"	"	"
E22	$6\overline{K}$	"	12	"	"	"	"
E23	$7\overline{K}$	"	40	"	"	"	"
E24	$8\overline{K}$	"	17	"	"	"	"

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NOTE R1 = 909Ω ±1%, 1/4 W.

* FIGURE 10. Input clamp diode voltage test circuit for table VII.

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TABLE VIII. Functional test.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$\overline{2Q}$	$\overline{3Q}$	$\overline{5Q}$	$\overline{6Q}$	$\overline{7Q}$	$\overline{8Q}$	Min	Max		
F1	Functional	2,3,4,6,7, 9,12,14,15, 16,17,19, 20,22,24, 25,27,28, 31,34,37, 39,40	30	1/ 2/	Open	H	H	H	H	H	H	H	H	L	L	L	L	L	L		3/		11
F2	"	2,3,4,6,7, 9,12,14,15, 16,17,20, 22,24,25, 27,28,30, 31,34,37, 39,40	19	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H		"		"
F3	"	2,3,4,6,7, 9,12,14,15, 16,17,19, 20,22,24, 25,27,28, 30,31,34, 37,39,40	Open	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H		"		"
F4	"	2,3,4,6,7, 9,12,14,15, 16,17,19, 20,22,24, 25,27,28, 31,34,37, 39,40	30	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L		"		"
F5	"	2,3,4,6,7, 9,12,14,15, 16,17,19, 20,22,24, 25,27,28, 30,31,34, 37,39,40	Open	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L		"		"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	2$\overline{Q}$	3$\overline{Q}$	5$\overline{Q}$	6$\overline{Q}$	7$\overline{Q}$	8$\overline{Q}$	Min	Max		
F6	Functional	2,3,4,6,7, 9,12,14,15, 16,17,20, 22,24,25, 27,28,31, 34,37,39, 40	19,30	1/ 2/	Open	H	H	H	H	H	H	H	H	H	H	H	H	H	H	3/			11
F7	"	2,3,4,6,7, 9,12,14, 15,16,17, 20,22,24, 25,27,28, 30,31,34, 37,39,40	19	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F8	"	2,3,4,6,7, 9,12,14,15, 16,17,20, 22,24,25, 27,28,31, 34,37,39, 40	19,30	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F9	"	2,3,4,6,7, 9,12,14,15, 16,17,19, 20,22,24, 25,27,28, 31,34,37, 39,40	30	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F10	"	2,3,6,9,12, 15,16,17, 22,25,28, 30,31,39, 40	4,7,14,19, 20,24,27, 34,37	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$2\overline{Q}$	$3\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	$7\overline{Q}$	$8\overline{Q}$	Min	Max		
F11	Functional	2,3,6,9,12, 15,16,17, 19,22,25, 28,30,31, 39,40	4,7,14,20, 24,27,34, 37	1/ 2/	Open	L	L	L	L	L	L	L	L	H	H	H	H	H	H	3/			11
F12	"	2,3,6,9,12, 15,16,17, 19,22,25, 28,31,39, 40	4,7,14,20, 24,27,30, 34,37	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F13	"	2,3,6,9,12, 15,16,17, 19,22,25, 28,30,31, 39,40	4,7,14,20, 24,27,34, 37	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F14	"	2,3,6,9,12, 15,16,17, 22,25,28, 31,39,40	4,7,14,19, 20,24,27, 30,34,37	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F15	"	2,3,6,9,12, 15,16,17, 22,25,28, 30,31,39, 40	4,7,14,19, 20,24,27, 34,37	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F16	"	2,3,6,9,12, 15,16,17, 22,25,28, 31,39,40	4,7,14,19, 20,24,27, 30,34,37	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F17	"	2,3,6,9,12, 15,16,17, 19,22,25, 28,31,39, 40	4,7,14,20, 24,27,30, 34,37	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	2$\overline{Q}$	3$\overline{Q}$	5$\overline{Q}$	6$\overline{Q}$	7$\overline{Q}$	8$\overline{Q}$	Min	Max		
F18	Functional	2,4,7,9,14, 16,20,22, 24,27,30, 31,34,37, 39	3,6,12,15, 17,19,25, 28,40	1/ 2/	Open	L	L	L	L	L	L	L	L	H	H	H	H	H	H	3/			11
F19	"	2,4,7,9,14, 16,19,20, 22,24,27, 30,31,34, 37,39	3,6,12,15, 17,25,28, 40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F20	"	2,4,7,9,14, 16,19,20, 22,24,27, 31,34,37, 39	3,6,12,15, 17,25,28, 30,40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F21	"	2,4,7,9,14, 16,19,20, 22,24,27, 30,31,34, 37,39	3,6,12,15, 17,25,28, 40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F22	"	2,4,7,9,14, 16,20,22, 24,27,31, 34,37,39	3,6,12,15, 17,19,25, 28,30,40	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F23	"	2,4,7,9,14, 16,20,22, 24,27,30, 31,34,37, 39	3,6,12,15, 17,19,25, 28,40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$2\overline{Q}$	$3\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	$7\overline{Q}$	$8\overline{Q}$	Min	Max		
F24	Functional	2,4,7,9,14, 16,20,22, 24,27,31, 34,37,39	3,6,12,15, 17,19,25, 28,30,40	1/ 2/	Open	H	H	H	H	H	H	H	H	H	H	H	H	H	H		3/		11
F25	"	2,4,7,9,14, 16,19,20, 22,24,27, 31,34,37, 39	3,6,12,15, 17,25,28, 30,40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L		"		"
F26	"	2,9,16,22, 30,31,39	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H		"		"
F27	"	2,9,16,19, 22,30,31, 39	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 34,37,40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H		"		"
F28	"	2,9,16,19, 22,31,39	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 30,34,37, 40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L		"		"
F29	"	2,9,16,19, 22,30,31, 39	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 34,37,40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L		"		"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$2\overline{Q}$	$3\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	$7\overline{Q}$	$8\overline{Q}$	Min	Max		
F30	Functional	2,9,16,22, 31,39	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,30,34, 37,40	1/ 2/	Open	H	H	H	H	H	H	H	H	H	H	H	H	H	H	3/			11
F31	"	2,9,16,22, 30,31,39	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F32	"	2,9,16,22, 31,39	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,30,34, 37,40	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F33	"	2,9,16,19, 22,31,39	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 30,34,37, 40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F34	"	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	2,9,16, 22,30,31, 39	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$2\overline{Q}$	$3\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	$7\overline{Q}$	$8\overline{Q}$	Min	Max		
F35	Functional	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 30,34,37, 40	2,9,16, 19,22,31, 39	1/ 2/	Open	L	L	L	L	L	L	L	L	H	H	H	H	H	H	3/			11
F36	"	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,30,34, 37,40	2,9,16, 22,31,39	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F37	"	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	2,9,16, 22,30,31, 39	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F38	"	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,30,34, 37,40	2,9,16, 22,31,39	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F39	"	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 34,37,40	2,9,16, 19,22,30, 31,39	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F40	"	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 30,34,37, 40	2,9,16, 19,22,31, 39	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$2\overline{Q}$	$3\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	$7\overline{Q}$	$8\overline{Q}$	Min	Max		
F41	Functional	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 34,37,40	2,9,16, 19,22,30, 31,39	1/ 2/	Open	H	H	H	H	H	H	H	H	H	H	H	H	H	H	3/			11
F42	"	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	2,9,16, 22,30,31, 39	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F43	"	3,6,12,15, 17,25,28, 30,40	2,4,7,9, 14,16,19, 20,22,24, 27,31,34, 37,39	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F44	"	3,6,12,15, 17,19,25, 28,30,40	2,4,7,9, 14,16,20, 22,24,27, 31,34,37, 38	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F45	"	3,6,12,15, 17,19,25, 28,40	2,4,7,9, 14,16,20, 22,24,27, 30,31,34, 37,39	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F46	"	3,6,12,15, 17,19,25, 28,30,40	2,4,7,9, 14,16,20, 22,24,27, 31,34,37, 39	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$2\overline{Q}$	$3\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	$7\overline{Q}$	$8\overline{Q}$	Min	Max		
F47	Functional	3,6,12,15, 17,25,28, 40	2,4,7,9, 14,16,19, 20,22,24, 27,30,31, 34,37,39	1/ 2/	Open	H	H	H	H	H	H	H	H	H	H	H	H	H	H	3/			11
F48	"	3,6,12,15, 17,25,28, 30,40	2,4,7,9, 14,16,19, 20,22,24, 27,31,34, 37,39	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F49	"	3,6,12,15, 17,25,28, 40	2,4,7,9, 14,16,19, 20,22,24, 27,30,31, 34,37,39	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F50	"	3,6,12,15, 17,19,25, 28,40	2,4,7,9, 14,16,20, 22,24,27, 30,31,34, 37,39	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F51	"	4,7,14,20, 24,27,30, 34,37	2,3,6,9, 12,15,16, 17,19,22, 25,28,31, 39,40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F52	"	4,7,14,19, 20,24,27, 30,34,37	2,3,6,9, 12,15,16, 17,22,25, 28,31,39, 40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$\overline{2Q}$	$\overline{3Q}$	$\overline{5Q}$	$\overline{6Q}$	$\overline{7Q}$	$\overline{8Q}$	Min	Max		
F53	Functional	4,7,14,19, 20,24,27, 34,37	2,3,6,9, 12,15,16, 17,22,25, 28,30,31, 39,40	1/ 2/	Open	H	H	H	H	H	H	H	H	L	L	L	L	L	L		3/		11
F54	"	4,7,14,19, 20,24,27, 30,34,37	2,3,6,9, 12,15,16, 17,22,25, 28,31,39, 40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L		"		"
F55	"	4,7,14,20, 24,27,34, 37	2,3,6,9, 12,15,16, 17,19,22, 25,28,30, 31,39,40	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H		"		"
F56	"	4,7,14,20, 24,27,30, 34,37	2,3,6,9, 12,15,16, 17,19,22, 25,28,31, 39,40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H		"		"
F57	"	4,7,14,20, 24,27,34, 37	2,3,6,9, 12,15,16, 17,19,22, 25,28,30, 31,39,40	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H		"		"
F58	"	4,7,14,19, 20,24,27, 34,37	2,3,6,9, 12,15,16, 17,22,25, 28,30,31, 39,40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L		"		"

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$2\overline{Q}$	$3\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	$7\overline{Q}$	$8\overline{Q}$	Min	Max		
F59	Functional	30	2,3,4,6, 7,9,12, 14,15,16, 17,19,20, 22,24,25, 27,28,31, 34,37,39, 40	1/ 2/	Open	L	L	L	L	L	L	L	L	H	H	H	H	H	H	3/			11
F60	"	19,30	2,3,4,6, 7,9,12, 14,15,16, 17,20,22, 24,25,27, 28,31,34, 37,39,40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F61	"	19	2,3,4,6, 7,9,12, 14,15,16, 17,20,22, 24,25,27, 28,30,31, 34,37,39, 40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F62	"	19,30	2,3,4,6, 7,9,12, 14,15,16, 17,20,22, 24,25,27, 28,31,34, 37,39,40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$2\overline{Q}$	$3\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	$7\overline{Q}$	$8\overline{Q}$	Min	Max		
F63	Functional	Open	2,3,4,6, 7,9,12, 14,15,16, 17,19,20, 22,24,25, 27,28,30, 31,34,37, 39,40	1/ 2/	Open	H	H	H	H	H	H	H	H	H	H	H	H	H	H	3/			11
F64	"	30	2,3,4,6, 7,9,12, 14,15,16, 17,19,20, 22,24,25, 27,28,31, 34,37,39, 40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F65	"	Open	2,3,4,6, 7,9,12, 14,15,16, 17,19,20, 22,24,25, 27,28,30, 31,34,37, 39,40	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F66	"	19	2,3,4,6, 7,9,12, 14,15,16, 17,20,22, 24,25,27, 28,30,31, 34,37,39, 40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"

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TABLE VIII. Functional test - Continued

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	2$\overline{Q}$	3$\overline{Q}$	5$\overline{Q}$	6$\overline{Q}$	7$\overline{Q}$	8$\overline{Q}$	Min	Max		
F67	Functional (input threshold check)	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	30	1/ 2/	2,9,16, 22,31, 39	H	H	H	H	H	H	H	H	L	L	L	L	L	L		3/		11,12
F68	"	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 30,34,37, 40	19	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H		"		"
F69	"	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,30,34, 37,40	Open	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H		"		"
F70	"	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 34,37,40	19,30	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H		"		"
F71	"	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,30,34, 37,40	Open	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H		"		"
F72	"	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	30	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L		"		"

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$2\overline{Q}$	$3\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	$7\overline{Q}$	$8\overline{Q}$	Min	Max		
F73	Functional (input threshold check)	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,30,34, 37,40	Open	1/ 2/	2,9,16, 22,31, 39	L	L	L	L	L	L	L	L	H	H	H	H	H	H	3/			11,12
F74	"	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 34,37,40	19,39	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F75	"	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	30	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F76	"	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 34,37,40	19,30	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F77	"	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	30	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F78	"	3,6,12, 15,17,25, 28,30,40	4,7,14, 19,20,24, 27,34,37	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	$2\overline{Q}$	$3\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	$7\overline{Q}$	$8\overline{Q}$	Min	Max		
F79	Functional (input threshold check)	3,6,12, 15,17,19, 25,28,30, 40	4,7,14, 20,24,27, 34,37	1/ 2/	2,9,16, 22,31, 39	H	H	H	H	H	H	H	H	L	L	L	L	L	L	3/			11,12
F80	"	3,6,12, 15,17,19, 25,28,30, 40	4,7,14, 20,24,27, 34,37	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F81	"	3,6,12, 15,17,19, 25,28,40	4,7,14, 20,24,27, 30,34,37	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F82	"	3,6,12, 15,17,25, 28,40	4,7,14, 19,20,24, 27,30,34, 37	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F83	"	3,6,12, 15,17,25, 28,30,40	4,7,14, 19,20,24, 27,34,37	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F84	"	3,6,12, 15,17,25, 28,40	4,7,14, 19,20,24, 27,30,34, 37	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F85	"	3,6,12, 15,17,19, 25,28,40	4,7,14, 20,24,27, 30,34,37	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F86	"	4,7,14, 20,24,27, 30,34,37	3,6,12, 15,17,19, 25,28,40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	2$\overline{Q}$	3$\overline{Q}$	5$\overline{Q}$	6$\overline{Q}$	7$\overline{Q}$	8$\overline{Q}$	Min	Max		
F87	Functional (input threshold check)	4,7,14, 19,20,24, 27,30,34, 37	3,6,12, 15,17,25, 28,40	1/ 2/	2,9,16, 22,31, 39	L	L	L	L	L	L	L	L	H	H	H	H	H	H	3/			11,12
F88	"	4,7,14, 19,20,24, 27,34,37	3,6,12, 15,17,25, 28,30,40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F89	"	4,7,14, 19,20,24, 27,30,34, 37	3,6,12, 15,17,25, 28,40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F90	"	4,7,14, 20,24,27, 34,37	3,6,12, 15,17,19, 25,28,30, 40	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F91	"	4,7,14, 20,24,27, 30,34,37	3,6,12, 15,17,19, 25,28,40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F92	"	4,7,14, 20,24,27, 34,37	3,6,12, 15,17,19, 25,28,30, 40	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F93	"	4,7,14, 19,20,24, 27,34,37	3,6,12, 15,17,25, 28,30,40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F94	"	30	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	2$\overline{Q}$	3$\overline{Q}$	5$\overline{Q}$	6$\overline{Q}$	7$\overline{Q}$	8$\overline{Q}$	Min	Max		
F95	Functional (input threshold check)	19,30	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 34,37,40	1/ 2/	2,9,16, 22,31, 39	H	H	H	H	H	H	H	H	L	L	L	L	L	L	3/			11,12
F96	"	Open	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,30,34, 37,40	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F97	"	19,30	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 34,37,40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"
F98	"	30	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"
F99	"	Open	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,30,34, 37,40	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F100	"	30	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,34,37, 40	"	"	L	L	L	L	L	L	L	L	H	H	H	H	H	H	"			"

See footnotes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect				Output logic states														Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	7Q	8Q	2$\overline{Q}$	3$\overline{Q}$	5$\overline{Q}$	6$\overline{Q}$	7$\overline{Q}$	8$\overline{Q}$	Min	Max		
F101	Functional (input threshold check)	19	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 30,34,37, 40	1/ 2/	2,9,16, 22,31, 39	H	H	H	H	H	H	H	H	L	L	L	L	L	L	3/			11,12
F102	"	Open	3,4,6,7, 12,14,15, 17,19,20, 24,25,27, 28,30,34, 37,40	"	"	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"			"
F103	"	19	3,4,6,7, 12,14,15, 17,20,24, 25,27,28, 30,34,37, 40	"	"	H	H	H	H	H	H	H	H	L	L	L	L	L	L	"			"

1/ For test conditions F1-F103, verification of the output logic states in note 3 is obtained by connecting item M02 (point C) to each output pin in succession after input conditions have been accomplished. Tests F1-F103 shall be performed in sequence.

2/ Q1, Q2, Q3, Q4, Q5, Q6, Q7, Q8, $\overline{Q}$2, $\overline{Q}$3, $\overline{Q}$5, $\overline{Q}$6, $\overline{Q}$7, and $\overline{Q}$8 output logic states are measured at output pins 5, 23, 29, 26, 13, 11, 18, 35, 21, 8, 33, 32, 38, and 36, respectively.

3/ Output logic states:
L = Low level = 0.0 to 0.4 V.
H = High level = 2.4 to 4.5 V.

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* TABLE IX. Output voltage (low-level and high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1Q	Output voltage, low level	30	19	5	Open	0	400	mV	11
G2	2Q	"	"	"	23	"	"	"	"	"
G3	3Q	"	"	"	29	"	"	"	"	"
G4	4Q	"	"	"	26	"	"	"	"	"
G5	5Q	"	"	"	13	"	"	"	"	"
G6	6Q	"	"	"	11	"	"	"	"	"
G7	7Q	"	"	"	18	"	"	"	"	"
G8	8Q	"	"	"	35	"	"	"	"	"
G9	$2\overline{Q}$	"	19	30	21	"	"	"	"	"
G10	$3\overline{Q}$	"	"	"	8	"	"	"	"	"
G11	$5\overline{Q}$	"	"	"	33	"	"	"	"	"
G12	$6\overline{Q}$	"	"	"	32	"	"	"	"	"
G13	$7\overline{Q}$	"	"	"	38	"	"	"	"	"
G14	$8\overline{Q}$	"	"	"	36	"	"	"	"	"

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* TABLE IX. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1Q	Output voltage, high level (input threshold check)	3,6,7, 12,14, 15,17, 19,20, 24,27, 28,30, 34,37, 40	4,25	5	2,9,16, 22,31, 39	2.4	4.5	V	11,12
H2	2Q	"	4,6,7, 12,14, 15,17, 19,20, 25,27, 28,30, 34,37, 40	3,24	23	"	"	"	"	"
H3	3Q	"	3,4,6, 12,14, 15,17, 19,20, 24,25, 27,30, 34,37, 40	7,28	29	"	"	"	"	"
H4	4Q	"	3,4,7, 12,14, 15,17, 19,20, 24,25, 28,30, 34,37, 40	6,27	26	"	"	"	"	"

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* TABLE IX. Output voltage (low-level and high-level) test - Continued.

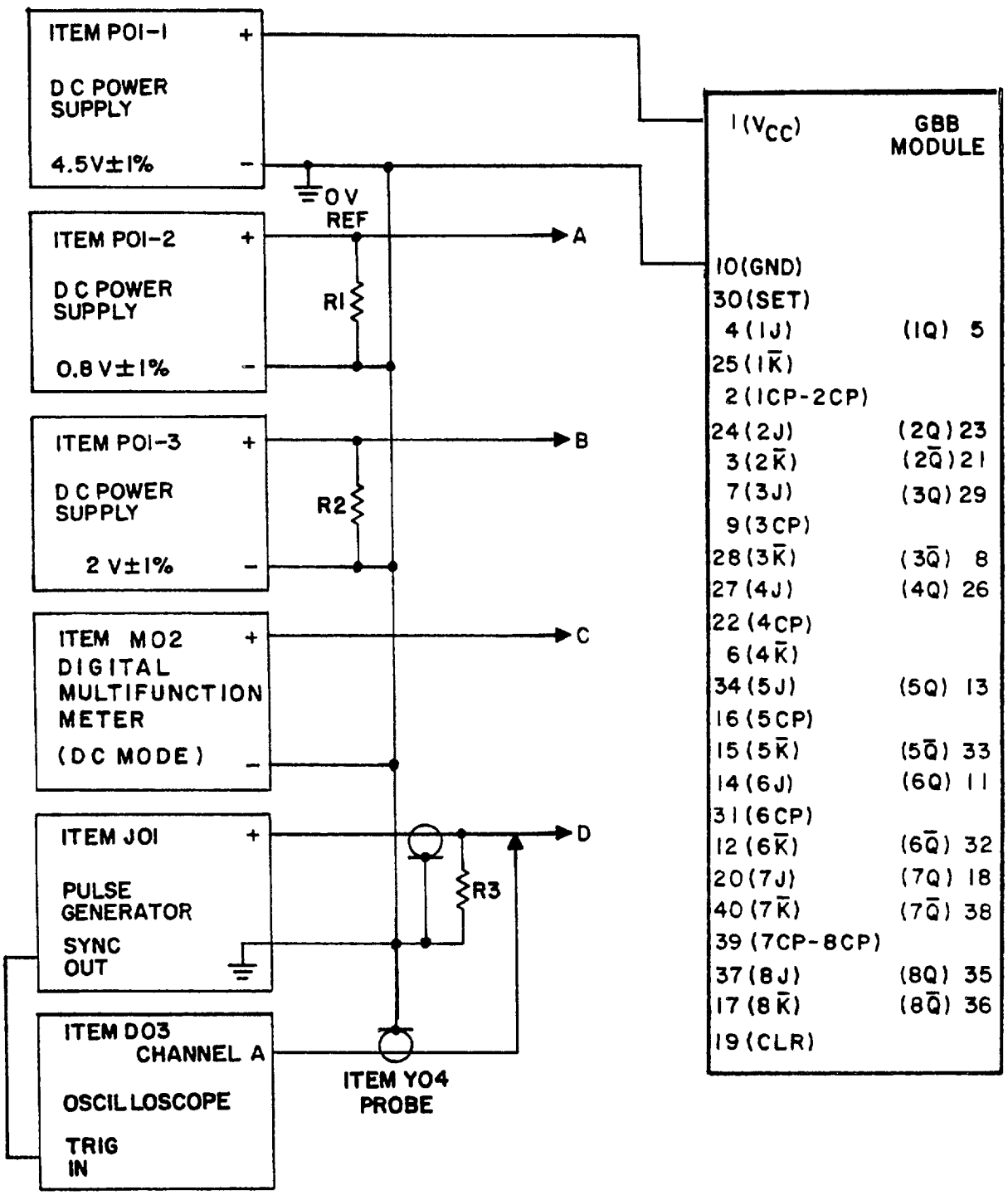
Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
H5	5Q	Output voltage, high level (input threshold check)	3,4,6, 7,12, 14,17, 19,20, 24,25, 27,28, 30,37, 40	15,34	13	2,9,16, 22,31, 39	2.4	4.5	V	11,12
H6	6Q	"	3,4,6, 7,15, 17,19, 20,24, 25,27, 28,30, 34,37, 40	12,14	11	"	"	"	"	"
H7	7Q	"	3,4,6, 7,12, 14,15, 19,24, 25,27, 28,30, 34,37	20,40	18	"	"	"	"	"
H8	8Q	"	3,4,6, 7,12, 14,15, 19,20, 24,25, 27,28, 30,34, 40	17,37	35	"	"	"	"	"
H9	$2\overline{Q}$	"	3,19, 24,30	4,6,7, 12,14, 15,17, 20,25, 27,28, 34,37, 40	21	"	"	"	"	"

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* TABLE IX. Output voltage (low-level and high-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
H10	$3\overline{Q}$	Output voltage, high level (input threshold check)	7,19, 28,30	3,4,6, 12,14, 15,17, 20,24, 25,27, 34,37, 40	8	2,9,16, 22,31, 39	2.4	4.5	V	11,12
H11	$5\overline{Q}$	"	15,19, 30,34	3,4,6, 7,12, 14,17, 20,24, 25,27, 28,37, 40	33	"	"	"	"	"
H12	$6\overline{Q}$	"	12,14, 19,30	3,4,6, 7,15, 17,20, 24,25, 27,28, 34,37, 40	32	"	"	"	"	"
H13	$7\overline{Q}$	"	19,20, 30,40	3,4,6, 7,12, 14,15, 17,24, 25,27, 28,34, 37	38	"	"	"	"	"
H14	$8\overline{Q}$	"	17,19, 30,37	3,4,6, 7,12, 14,15, 20,24, 25,27, 28,34, 40	36	"	"	"	"	"

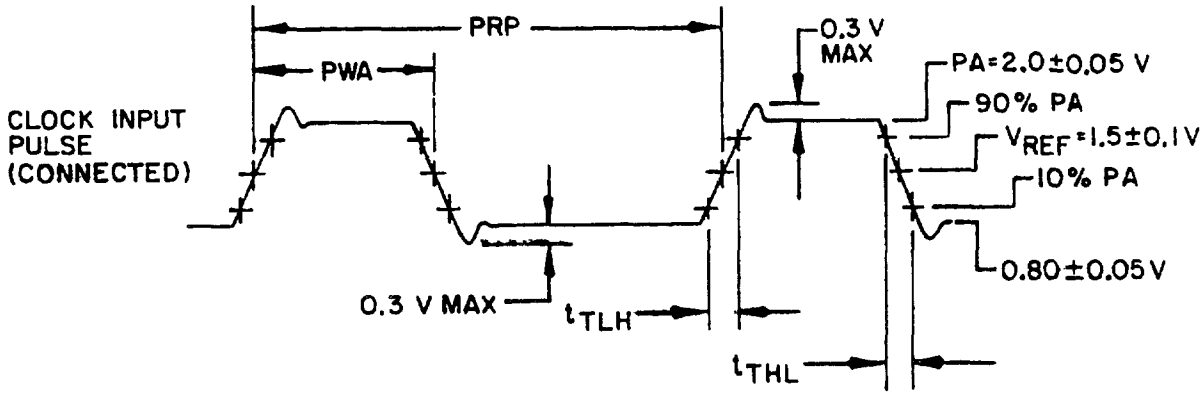
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 51.1Ω ±1%, 1/4 W.
3. For test conditions F67-F103 of table VIII and H1-H14 of table IX, make all input circuit connections for a test condition then use the single pulse mode on item J01 (after initial setup) and apply a single pulse before making each measurement.
4. Test conditions F1-F103 shall be performed in sequence.
5. For test conditions G1-G14 and H1-H14, load the output under test and the corresponding complement output (where available) with the load circuit as shown on figure 13. Load circuits are not required for test conditions F1-F103.

* FIGURE 11. Functional and output voltage (low-level and high-level) test circuit for tables VIII and IX.

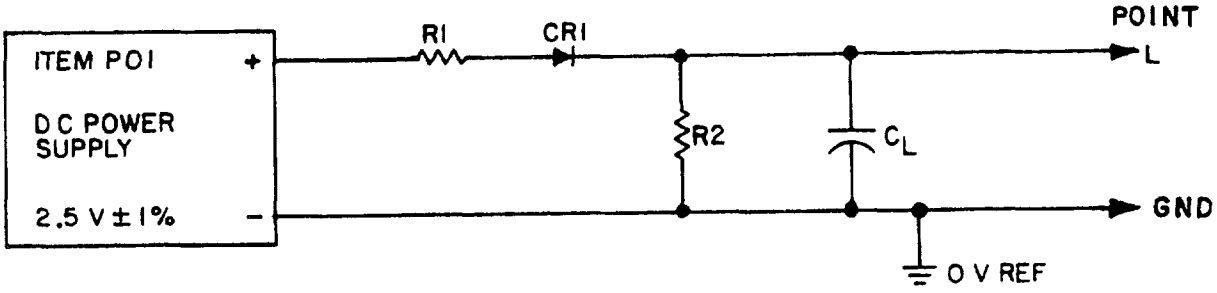
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NOTES:
1. Clock input pulse parameters:
   PWA = 500 ±100 ns.
   PRP = 1,000 ±200 ns.
   $t_{TLH} = t_{THL}$ = 50 ±25 ns.
2. Clock threshold level setup:
   With the clock input connected, place item J01 in the single pulse mode. Use item D03 to adjust dc offset of item J01 for a low level of 0.80 ±0.05 V (clock low level threshold). Adjust item J01 to obtain the required PWA and PRP. Use item D03 to adjust item J01 amplitude for a high level of 2.00 ±0.05 V (clock high level threshold).

FIGURE 12. Functional and output voltage (high-level) test circuit waveform for tables VIII and IX.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 16.0 mA ±5% out of point L with point L held at 0.4 V.
3. R2 = 2.94 kΩ ±1%, 1/8 W.
4. Test type S:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
   Test type Q:
   C_L = 150 ±5 pF (including probe and parasitic capacitance).
5. Capacitor C_L is not required for dc electrical test conditions G1-G14 and H1-H14.
6. Lead length from load circuit to module output pin under test shall be less than 2 inches (50.8 mm).

* FIGURE 13. GBB logic load circuit.

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TABLE X. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	CLR Input current, high level	19	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40	0	160	µA	14
J2	1CP-2CP "	2	3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40	"	"	"	"
J3	7CP-8CP "	39	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,40	"	"	"	"
J4	SET "	30	2,3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,31,34, 37,39,40	"	80	"	"
J5	3CP "	9	2,3,4,6,7,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40	"	"	"	"
J6	4CP "	22	2,3,4,6,7,9,12,14,15,16,17, 20,24,25,27,28,30,31,34,37, 39,40	"	"	"	"
J7	5CP "	16	2,3,4,6,7,9,12,14,15,17,20, 22,24,25,27,28,30,31,34,37, 39,40	"	"	"	"
J8	6CP "	31	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,34,37, 39,40	"	"	"	"
J9	1J "	4	2,3,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40	"	40	"	"
J10	2J "	24	2,3,4,6,7,9,12,14,15,16,17, 20,22,25,27,28,30,31,34,37, 39,40	"	40	"	"

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TABLE X. Input current (high-level) test - Continued.

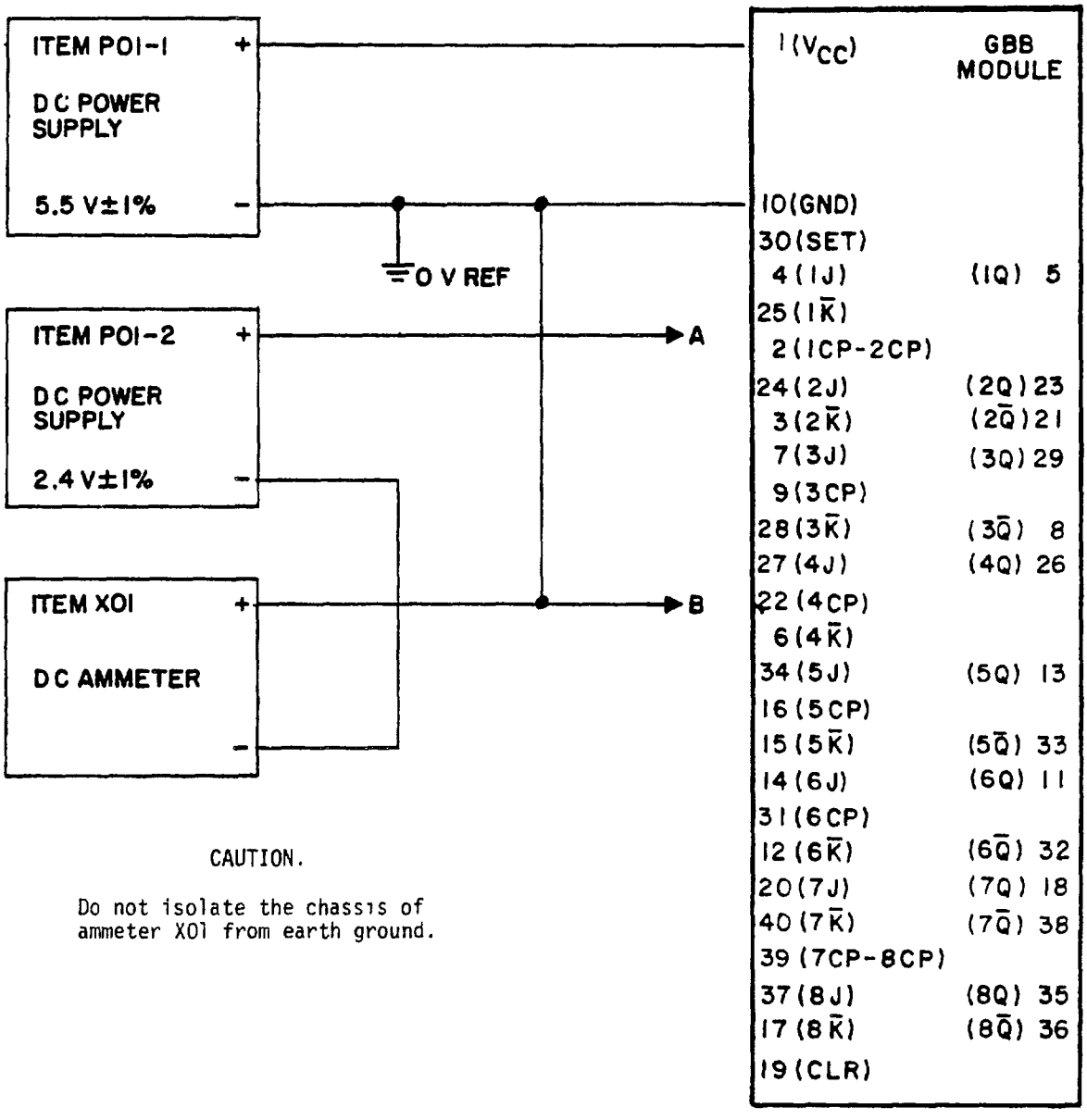
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J11	3J	Input current, high level	7	2,3,4,6,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40	0	40	µA	14
J12	4J	"	27	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,28,30,31,34,37, 39,40	"	"	"	"
J13	5J	"	34	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,37, 39,40	"	"	"	"
J14	6J	"	14	2,3,4,6,7,9,12,15,16,17,20, 22,24,25,27,28,30,31,34,37, 39,40	"	"	"	"
J15	7J	"	20	2,3,4,6,7,9,12,14,15,16,17, 22,24,25,27,28,30,31,34,37, 39,40	"	"	"	"
J16	8J	"	37	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 39,40	"	"	"	"
J17	1$\overline{K}$	"	25	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,27,28,30,31,34,37, 39,40	"	"	"	"
J18	2$\overline{K}$	"	3	2,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40	"	"	"	"
J19	3$\overline{K}$	"	28	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,30,31,34,37, 39,40	"	"	"	"
J20	4$\overline{K}$	"	6	2,3,4,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39,40	"	"	"	"

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TABLE X. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J21	$5\overline{K}$	Input current, high level	15	2,3,4,6,7,9,12,14,16,17,20, 22,24,25,27,28,30,31,34,37, 39,40	0	40	µA	14
J22	$6\overline{K}$	"	12	2,3,4,6,7,9,14,15,16,17,20, 22,24,25,27,28,30,31,34,37, 39,40	"	"	"	"
J23	$7\overline{K}$	"	40	2,3,4,6,7,9,12,14,15,16,17, 20,22,24,25,27,28,30,31,34, 37,39	"	"	"	"
J24	$8\overline{K}$	"	17	2,3,4,6,7,9,12,14,15,16,20, 22,24,25,27,28,30,31,34,37, 39,40	"	"	"	"
J25	1CP-2CP	"	2	3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,31,34, 37,39,40	"	160	"	"
J26	7CP-8CP	"	39	2,3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,31,34, 37,40	"	160	"	"
J27	3CP	"	9	2,3,4,6,7,12,14,15,16,17, 19,20,22,24,25,27,28,31,34, 37,39,40	"	80	"	"
J28	4CP	"	22	2,3,4,6,7,9,12,14,15,16,17, 19,20,24,25,27,28,31,34,37, 39,40	"	"	"	"
J29	5CP	"	16	2,3,4,6,7,9,12,14,15,17,19, 20,22,24,25,27,28,31,34,37, 39,40	"	"	"	"
J30	6CP	"	31	2,3,4,6,7,9,12,14,15,16,17, 19,20,22,24,25,27,28,34,37, 39,40	"	"	"	"

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CAUTION.

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 14. Input current (high-level) test circuit for table X.

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TABLE XI. Input current (low-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	CLR Input current, low level	Open	19	-0.40	-6.92	mA	15
K2	1CP-2CP "	3,4,19,24,25	2	"	"	"	"
K3	7CP-8CP "	17,19,20,37, 40	39	"	"	"	"
K4	SET "	Open	30	-0.20	-3.46	"	"
K5	3CP "	7,19,28	9	"	"	"	"
K6	4CP "	6,19,27	22	"	"	"	"
K7	5CP "	15,19,34	16	"	"	"	"
K8	6CP "	12,14,19	31	"	"	"	"
K9	1J "	19,30	4	-0.10	-1.73	"	"
K10	2J "	"	24	"	"	"	"
K11	3J "	"	7	"	"	"	"
K12	4J "	"	27	"	"	"	"
K13	5J "	"	34	"	"	"	"
K14	6J "	"	14	"	"	"	"
K15	7J "	"	20	"	"	"	"
K16	8J "	"	37	"	"	"	"
K17	$1\overline{K}$ "	19	25	"	"	"	"
K18	$2\overline{K}$ "	19	3	"	"	"	"

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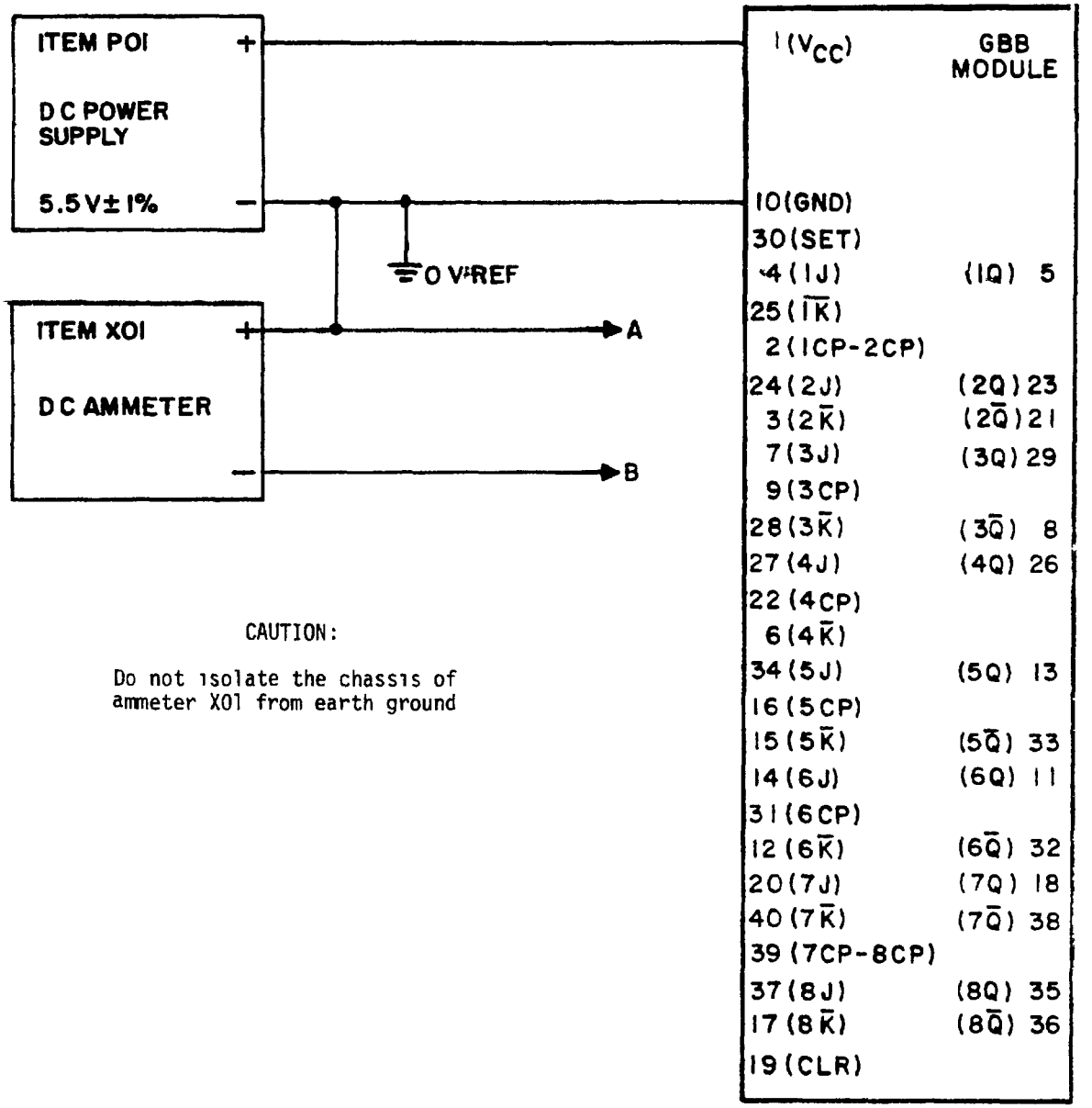
TABLE XI. Input current (low-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K19	$3\overline{K}$	Input current, low level	19	28	-0.10	-1.73	mA	15
K20	$4\overline{K}$	"	"	6	"	"	"	"
K21	$5\overline{K}$	"	"	15	"	"	"	"
K22	$6\overline{K}$	"	"	12	"	"	"	"
K23	$7\overline{K}$	"	"	40	"	"	"	"
K24	$8\overline{K}$	"	"	17	"	"	"	"

TABLE XII. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
L	Power supply current	30	144	mA	16

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground

FIGURE 15. Input current (low-level) test circuit for table XI.

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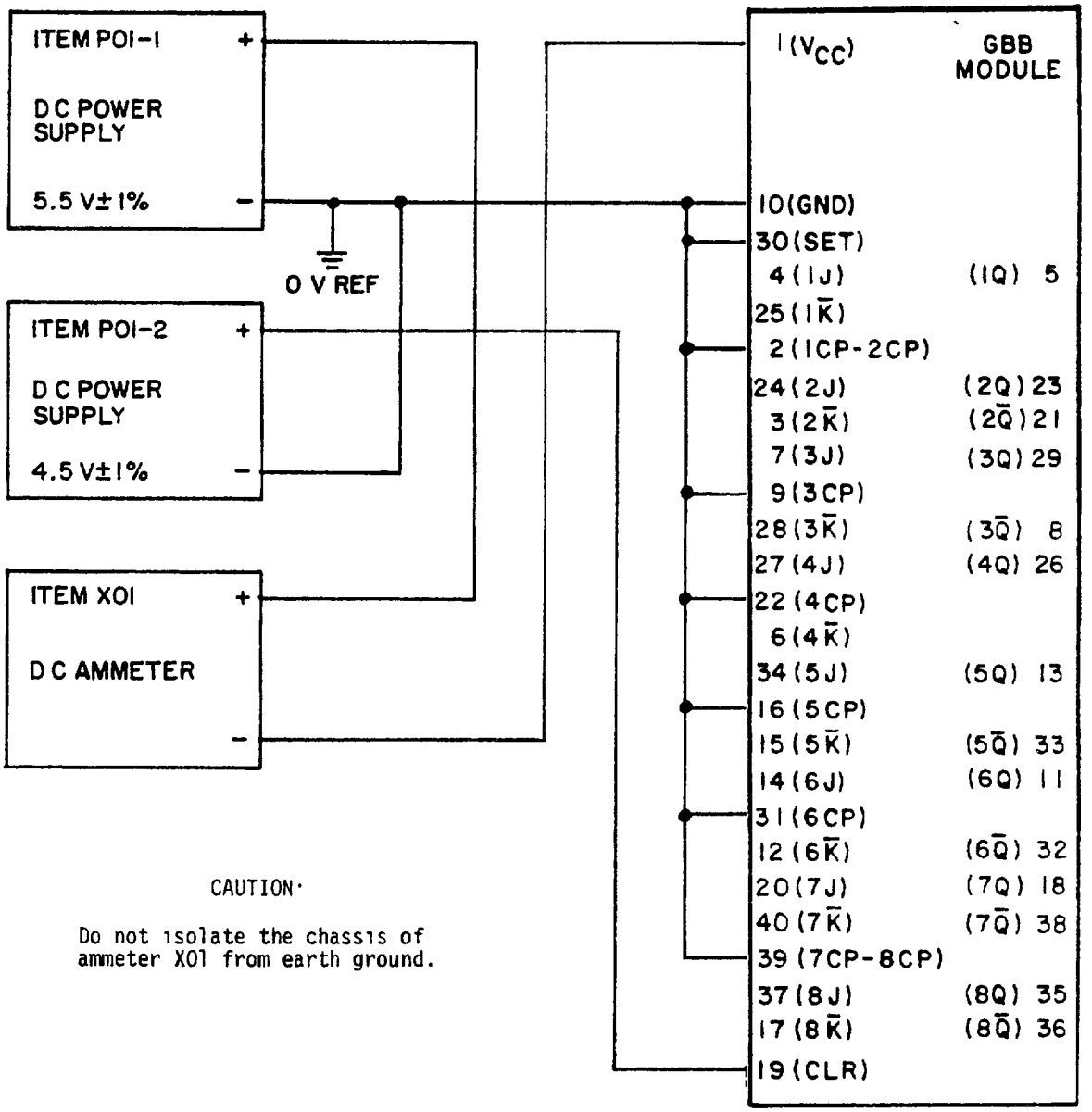


FIGURE 16. Power supply current test circuit for table XII.

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) (setup and hold time check) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	1Q Transition time (t_{TLH}), C_L = 50 pF	4,25	2	5	Open	4	30	ns	17,18
M2	2Q "	3,24	2	23	21	"	"	"	"
M3	3Q "	7,28	9	29	8	"	"	"	"
M4	4Q "	6,27	22	26	Open	"	"	"	"
M5	5Q "	15,34	16	13	33	"	"	"	"
M6	6Q "	12,14	31	11	32	"	"	"	"
M7	7Q "	20,40	39	18	38	"	"	"	"
M8	8Q "	17,37	39	35	36	"	"	"	"
M9	2$\overline{Q}$ "	3,24	2	21	23	"	"	"	"
M10	3$\overline{Q}$ "	7,28	9	8	29	"	"	"	"
M11	5$\overline{Q}$ "	15,34	16	33	13	"	"	"	"
M12	6$\overline{Q}$ "	12,14	31	32	11	"	"	"	"
M13	7$\overline{Q}$ "	20,40	39	38	18	"	"	"	"
M14	8$\overline{Q}$ "	17,37	39	36	35	"	"	"	"
M15	1Q Transition time (t_{TLH}), C_L = 150 pF	4,25	2	5	Open	"	50	"	"
M16	2Q "	3,24	2	23	21	"	"	"	"
M17	3Q "	7,28	9	29	8	"	"	"	"
M18	4Q "	6,27	22	26	Open	"	"	"	"
M19	5Q "	15,34	16	13	33	"	"	"	"

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) (setup and hold time check) test - Continued.

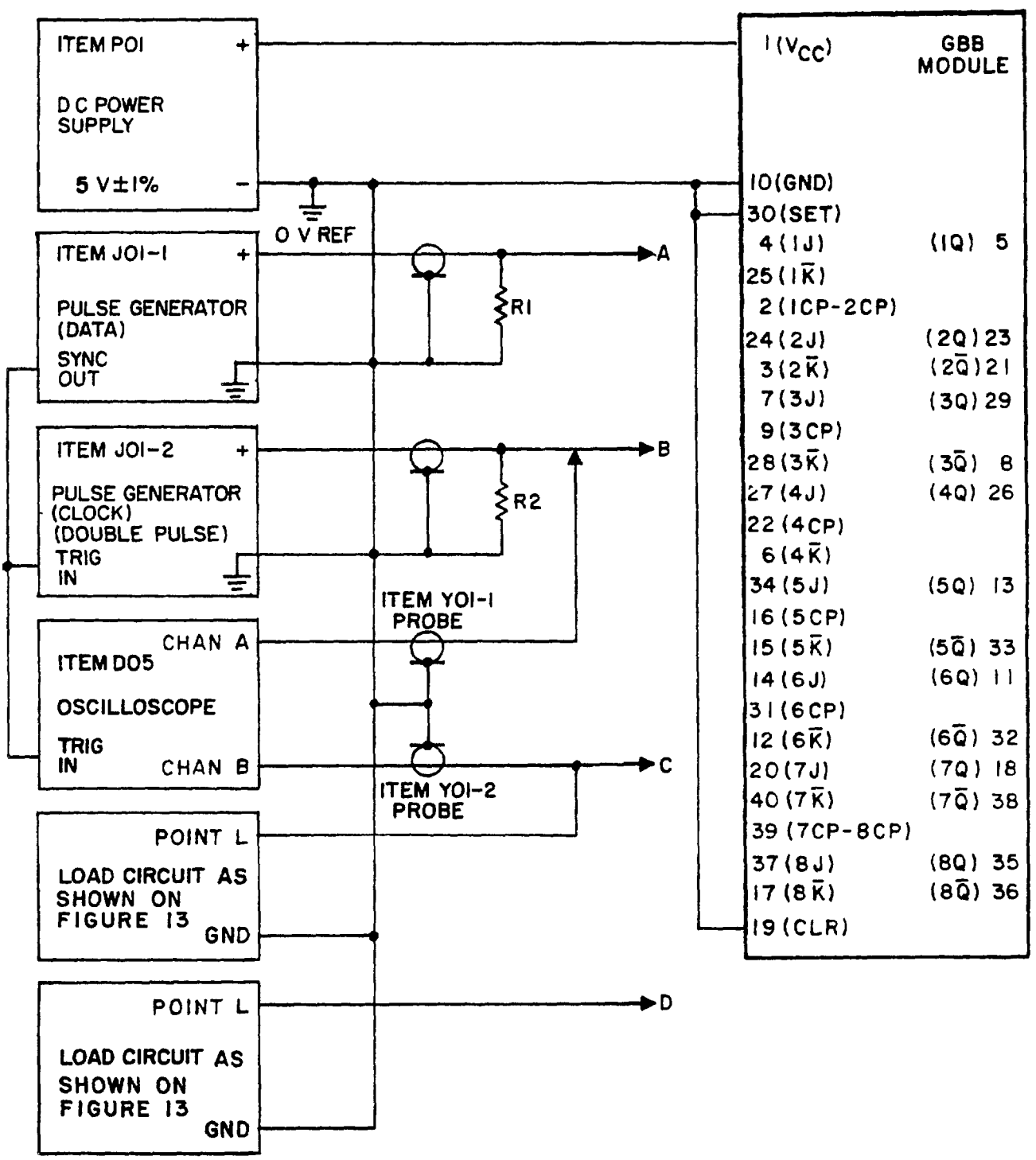
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M20	6Q Transition time (t_{TLH}), C_L = 150 pF	12,14	31	11	32	4	50	ns	17,18
M21	7Q "	20,40	39	18	38	"	"	"	"
M22	8Q "	17,37	39	35	36	"	"	"	"
M23	2$\overline{Q}$ "	3,24	2	21	23	"	"	"	"
M24	3$\overline{Q}$ "	7,28	9	8	29	"	"	"	"
M25	5$\overline{Q}$ "	15,34	16	33	13	"	"	"	"
M26	6$\overline{Q}$ "	12,14	31	32	11	"	"	"	"
M27	7$\overline{Q}$ "	20,40	39	38	18	"	"	"	"
M28	8$\overline{Q}$ "	17,37	39	36	35	"	"	"	"
N1	1Q Transition time (t_{THL}), C_L = 50 pF	4,25	2	5	Open	"	13	"	"
N2	2Q "	3,24	2	23	21	"	"	"	"
N3	3Q "	7,28	9	29	8	"	"	"	"
N4	4Q "	6,27	22	26	Open	"	"	"	"
N5	5Q "	15,34	16	13	33	"	"	"	"
N6	6Q "	12,14	31	11	32	"	"	"	"
N7	7Q "	20,40	39	18	38	"	"	"	"
N8	8Q "	17,37	39	35	36	"	"	"	"
N9	2$\overline{Q}$ "	3,24	2	21	23	"	"	"	"
N10	3$\overline{Q}$ "	7,28	9	8	29	"	"	"	"

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) (setup and hold time check) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N11	$5\overline{Q}$ Transition time (t_{THL}), C_L = 50 pF	15,34	16	33	13	4	13	ns	17,18
N12	$6\overline{Q}$ "	12,14	31	32	11	"	"	"	"
N13	$7\overline{Q}$ "	20,40	39	38	18	"	"	"	"
N14	$8\overline{Q}$ "	17,37	39	36	35	"	"	"	"
N15	1Q Transition time (t_{THL}), C_L = 150 pF	4,25	2	5	Open	"	20	"	"
N16	2Q "	3,24	2	23	21	"	"	"	"
N17	3Q "	7,28	9	29	8	"	"	"	"
N18	4Q "	6,27	22	26	Open	"	"	"	"
N19	5Q "	15,34	16	13	33	"	"	"	"
N20	6Q "	12,14	31	11	32	"	"	"	"
N21	7Q "	20,40	39	18	38	"	"	"	"
N22	8Q "	17,37	39	35	36	"	"	"	"
N23	$2\overline{Q}$ "	3,24	2	21	23	"	"	"	"
N24	$3\overline{Q}$ "	7,28	9	8	29	"	"	"	"
N25	$5\overline{Q}$ "	15,34	16	33	13	"	"	"	"
N26	$6\overline{Q}$ "	12,14	31	32	11	"	"	"	"
N27	$7\overline{Q}$ "	20,40	39	38	18	"	"	"	"
N28	$8\overline{Q}$ "	17,37	39	36	35	"	"	"	"

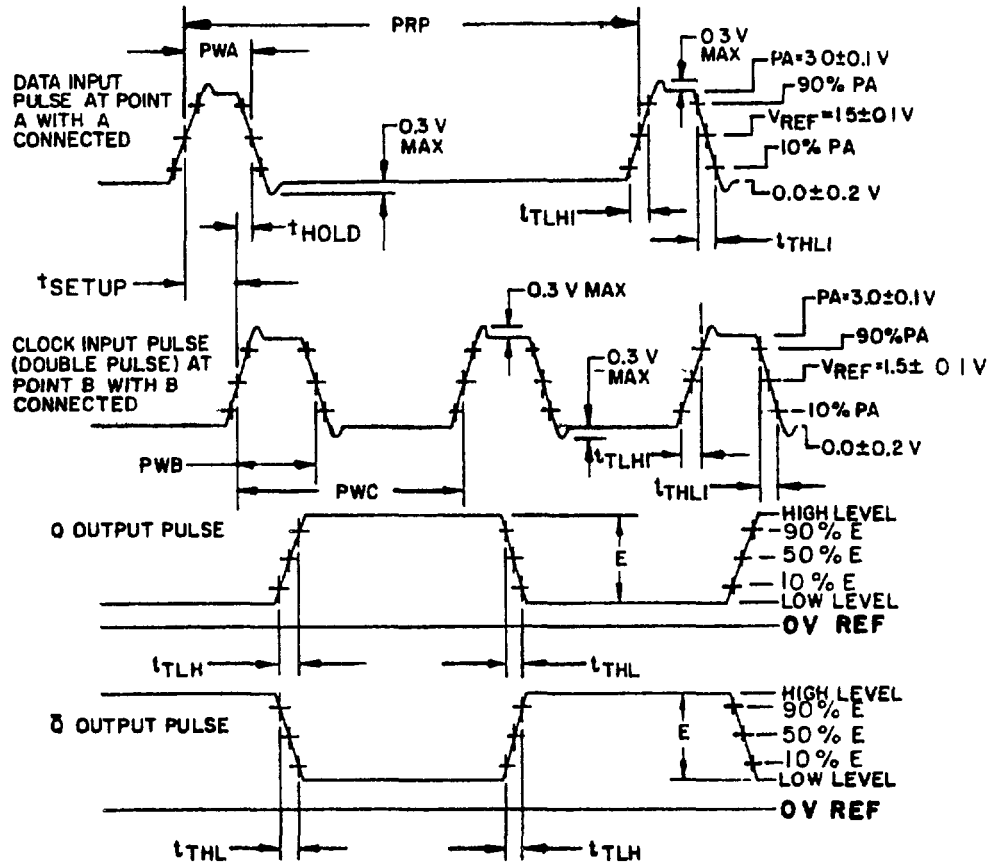
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NOTE: R1 = R2 = 51.1Ω ±1%, 1/4 W.

* FIGURE 17. Transition time (t_{TLH} and t_{THL}) (setup and hold time check) test circuit for table XIII.

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NOTES:

1. Data input pulse parameters:

   PRP = 1,000 ±200 ns.
   $t_{TLH1} = t_{THL1}$ = 6 ±1 ns.
   PWA = See note 3.

2. Clock input pulse parameters:

   Item J01-2 is operated in double pulse mode.
   PWB = 200 ±20 ns.
   $t_{TLH1} = t_{THL1}$ = 6 ±1 ns.
   Use delay control of item J01-2 to obtain PWC = 500 ±50 ns.

3. Minimum setup and hold time check:

   Use pulse width and pulse delay controls on item J01-1 ot obtain t_{SETUP} = 20 ±1 ns (minimum setup time) and t_{HOLD} = 6.0 ±0.5 ns (minimum hold time).
   PWA = 26 ns ($t_{SETUP} + t_{HOLD}$).
   The setup and hold time requirements must be verified before making measurements.

4. Verify that all outputs have a high level ≥ 2.4 V and a low level ≤ 0.4 V before making measurements per table XIII.
5. All voltages are referenced to ground.

FIGURE 18. Transition time (t_{TLH} and t_{THL}) (setup and hold time check) test circuit waveforms for table XIII.

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TABLE XIV. Propagation delay time from clock (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P1	1Q Propagation delay time from clock (t_{PLH}), C_L = 50 pF	4	25	2	5	Open	6	30	ns	19,20
P2	2Q "	24	3	2	23	21	"	"	"	"
P3	3Q "	7	28	9	29	8	"	"	"	"
P4	4Q "	27	6	22	26	Open	"	"	"	"
P5	5Q "	34	15	16	13	33	"	"	"	"
P6	6Q "	14	12	31	11	32	"	"	"	"
P7	7Q "	20	40	39	18	38	"	"	"	"
P8	8Q "	37	17	39	35	36	"	"	"	"
P9	2$\overline{Q}$ "	24	3	2	21	23	"	"	"	"
P10	3$\overline{Q}$ "	7	28	9	8	29	"	"	"	"
P11	5$\overline{Q}$ "	34	15	16	33	13	"	"	"	"
P12	6$\overline{Q}$ "	14	12	31	32	11	"	"	"	"
P13	7$\overline{Q}$ "	20	40	39	38	18	"	"	"	"
P14	8$\overline{Q}$ "	37	17	39	36	35	"	"	"	"
P15	1Q Propagation delay time from clock (t_{PLH}), C_L = 150 pF	4	25	2	5	Open	9	37	"	"
P16	2Q "	24	3	2	23	21	"	"	"	"
P17	3Q "	7	28	9	29	8	"	"	"	"
P18	4Q "	27	6	22	26	Open	"	"	"	"
P19	5Q "	34	15	16	13	33	"	"	"	"

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TABLE XIV. Propagation delay time from clock (t_{PLH} and t_{PHL}) test - Continued.

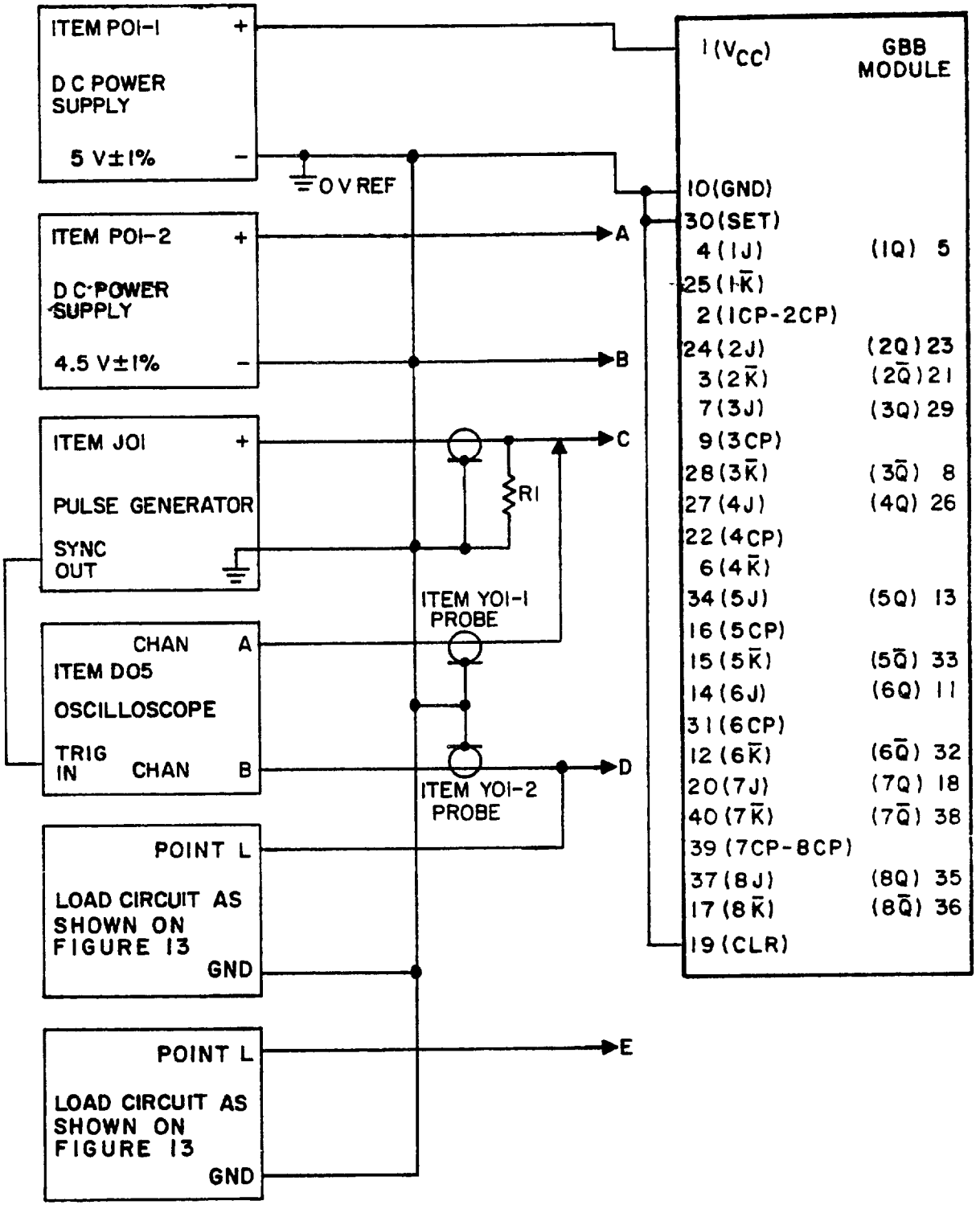
Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P20	6Q	Propagation delay time from clock (t_{PLH}), C_L = 150 pF	14	12	31	11	32	9	37	ns	19,20
P21	7Q	"	20	40	39	18	38	"	"	"	"
P22	8Q	"	37	17	39	35	36	"	"	"	"
P23	$2\overline{Q}$	"	24	3	2	21	23	"	"	"	"
P24	$3\overline{Q}$	"	7	28	9	8	29	"	"	"	"
P25	$5\overline{Q}$	"	34	15	16	33	13	"	"	"	"
P26	$6\overline{Q}$	"	14	12	31	32	11	"	"	"	"
P27	$7\overline{Q}$	"	20	40	39	38	18	"	"	"	"
P28	$8\overline{Q}$	"	37	17	39	36	35	"	"	"	"
Q1	1Q	Propagation delay time from clock (t_{PHL}), C_L = 50 pF	4	25	2	5	Open	8	40	"	"
Q2	2Q	"	24	3	2	23	21	"	"	"	"
Q3	3Q	"	7	28	9	29	8	"	"	"	"
Q4	4Q	"	27	6	22	26	Open	"	"	"	"
Q5	5Q	"	34	15	16	13	33	"	"	"	"
Q6	6Q	"	14	12	31	11	32	"	"	"	"
Q7	7Q	"	20	40	39	18	38	"	"	"	"
Q8	8Q	"	37	17	39	35	36	"	"	"	"
Q9	$2\overline{Q}$	"	24	3	2	21	23	"	"	"	"
Q10	$3\overline{Q}$	"	7	28	9	8	29	"	"	"	"

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TABLE XIV. Propagation delay time from clock (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q11	$5\overline{Q}$ Propagation delay time from clock (t_{PHL}), C_L = 50 pF	34	15	16	33	13	8	40	ns	19,20
Q12	$6\overline{Q}$ "	14	12	31	32	11	"	"	"	"
Q13	$7\overline{Q}$ "	20	40	39	38	18	"	"	"	"
Q14	$8\overline{Q}$ "	37	17	39	36	35	"	"	"	"
Q15	1Q Propagation delay time from clock (t_{PHL}), C_L = 150 pF	4	25	2	5	Open	10	58	"	"
Q16	2Q "	24	3	2	23	21	"	"	"	"
Q17	3Q "	7	28	9	29	8	"	"	"	"
Q18	4Q "	27	6	22	26	Open	"	"	"	"
Q19	5Q "	34	15	16	13	33	"	"	"	"
Q20	6Q "	14	12	31	11	32	"	"	"	"
Q21	7Q "	20	40	39	18	38	"	"	"	"
Q22	8Q "	37	17	39	35	36	"	"	"	"
Q23	$2\overline{Q}$ "	24	3	2	21	23	"	"	"	"
Q24	$3\overline{Q}$ "	7	28	9	8	29	"	"	"	"
Q25	$5\overline{Q}$ "	34	15	16	33	13	"	"	"	"
Q26	$6\overline{Q}$ "	14	12	31	32	11	"	"	"	"
Q27	$7\overline{Q}$ "	20	40	39	38	18	"	"	"	"
Q28	$8\overline{Q}$ "	37	17	39	36	35	"	"	"	"

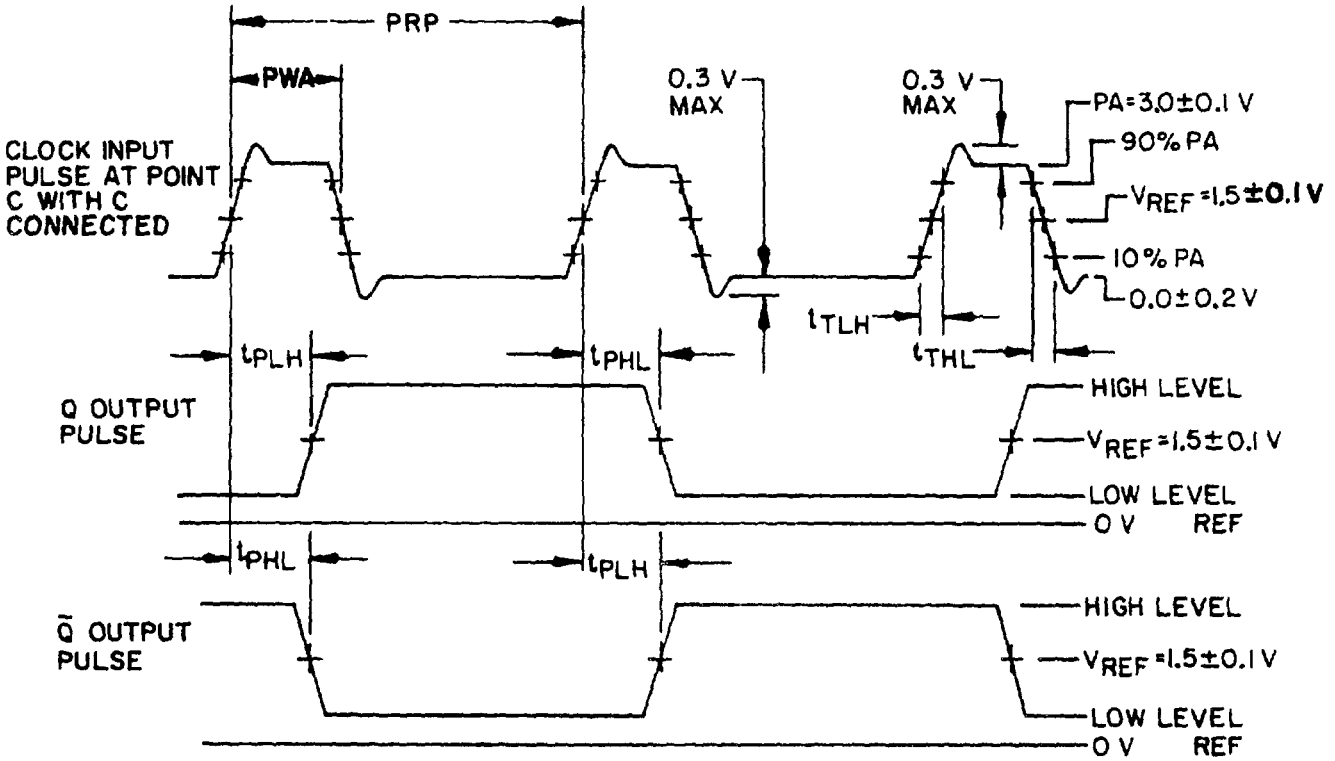
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

* FIGURE 19. Propagation delay time from clock (t_{PLH} and t_{PHL}) test circuit for table XIV.

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NOTES:
1. Clock input pulse parameters:
   PRP = 1,000 ±200 ns
   PWA = 250 ±50 ns
   $t_{TLH} = t_{THL}$ = 6 ±1 ns.
2. Verify that all outputs have a high level ≥ 2.4 V and a low level ≤ 0.4 V before making measurements.
3. All voltages are referenced to ground.

FIGURE 20. Propagation delay time from clock (t_{PLH} and t_{PHL}) test circuit waveforms for table XIV.

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TABLE XV. Propagation delay time from set and clear (t_{PLH} and t_{PHL}) (minimum set and clear pulse width check) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1Q	Propagation delay time from set (t_{PLH}), C_L = 50 pF	30	19	5	Open	8	35	ns	21,22
R2	2Q	"	"	"	23	21	"	"	"	"
R3	3Q	"	"	"	29	8	"	"	"	"
R4	4Q	"	"	"	26	Open	"	"	"	"
R5	5Q	"	"	"	13	33	"	"	"	"
R6	6Q	"	"	"	11	32	"	"	"	"
R7	7Q	"	"	"	18	38	"	"	"	"
R8	8Q	"	"	"	35	36	"	"	"	"
R9	1Q	Propagation delay time from set (t_{PLH}), C_L = 150 pF	"	"	5	Open	"	42	"	"
R10	2Q	"	"	"	23	21	"	"	"	"
R11	3Q	"	"	"	29	8	"	"	"	"
R12	4Q	"	"	"	26	Open	"	"	"	"
R13	5Q	"	"	"	13	33	"	"	"	"
R14	6Q	"	"	"	11	32	"	"	"	"
R15	7Q	"	"	"	18	38	"	"	"	"
R16	8Q	"	"	"	35	36	"	"	"	"
S1	$2\overline{Q}$	Propagation delay time from set (t_{PHL}), C_L = 50 pF	30	19	21	23	15	64	"	"
S2	$3\overline{Q}$	"	"	"	8	29	"	"	"	"

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TABLE XV. Propagation delay time from set and clear (t_{PLH} and t_{PHL}) (minimum set and clear pulse width check) test - Continued.

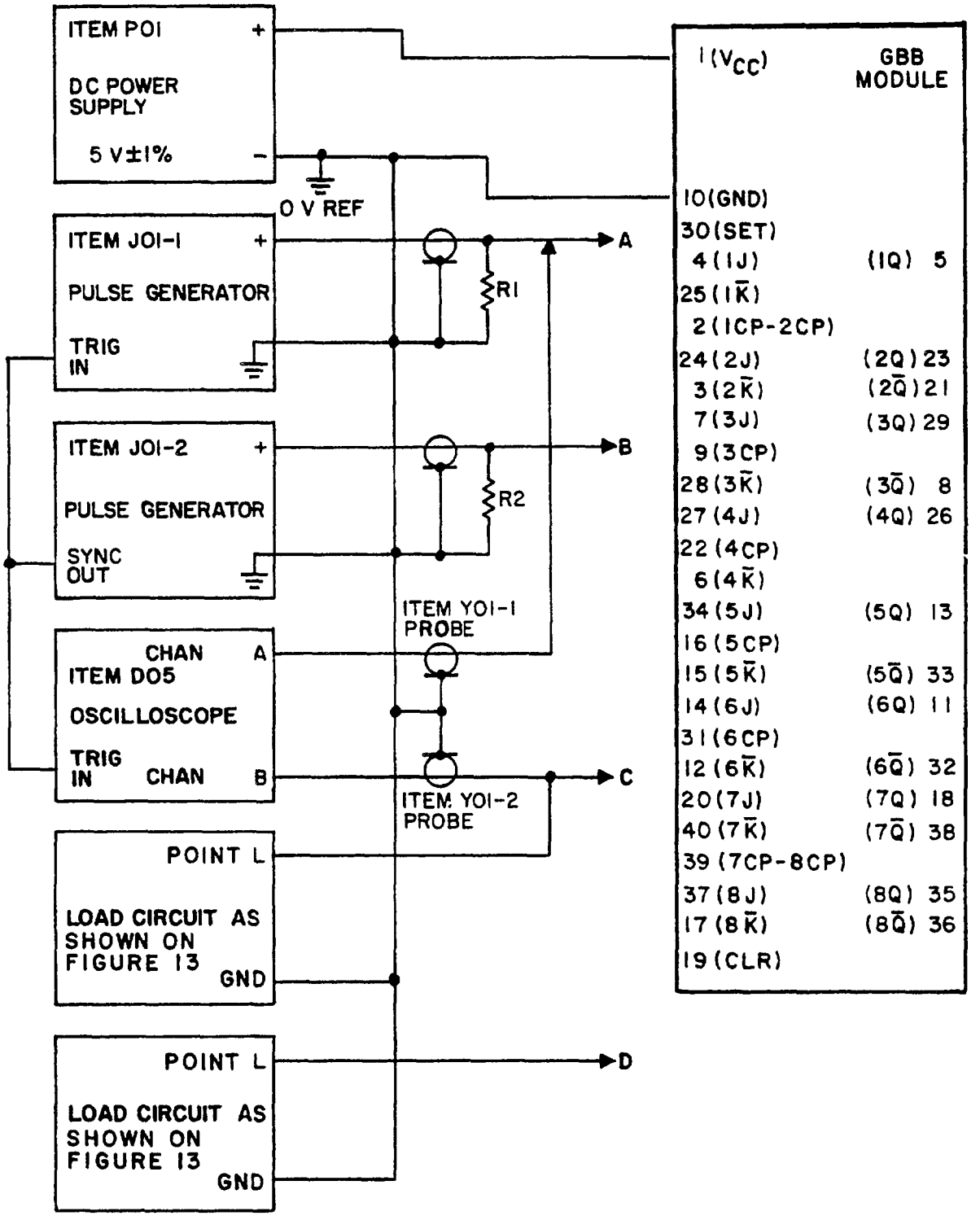
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S3	$5\overline{Q}$ Propagation delay time from set (t_{PHL}), C_L = 50 pF	30	19	33	13	15	64	ns	21,22
S4	$6\overline{Q}$ "	"	"	32	11	"	"	"	"
S5	$7\overline{Q}$ "	"	"	38	18	"	"	"	"
S6	$8\overline{Q}$ "	"	"	36	35	"	"	"	"
S7	$2\overline{Q}$ Propagation delay time from set (t_{PHL}), C_L = 150 pF	"	"	21	23	18	75	"	"
S8	$3\overline{Q}$ "	"	"	8	29	"	"	"	"
S9	$5\overline{Q}$ "	"	"	33	13	"	"	"	"
S10	$6\overline{Q}$ "	"	"	32	11	"	"	"	"
S11	$7\overline{Q}$ "	"	"	38	18	"	"	"	"
S12	$8\overline{Q}$ "	"	"	36	35	"	"	"	"
T1	$2\overline{Q}$ Propagation delay time from clear (t_{PLH}), C_L = 50 pF	19	30	21	23	8	35	"	21,23
T2	$3\overline{Q}$ "	"	"	8	29	"	"	"	"
T3	$5\overline{Q}$ "	"	"	33	13	"	"	"	"
T4	$6\overline{Q}$ "	"	"	32	11	"	"	"	"
T5	$7\overline{Q}$ "	"	"	38	18	"	"	"	"
T6	$8\overline{Q}$ "	"	"	36	35	"	"	"	"
T7	$2\overline{Q}$ Propagation delay time from clear (t_{PLH}), C_L = 150 pF	"	"	21	23	"	42	"	"
T8	$3\overline{Q}$ "	"	"	8	29	"	"	"	"
T9	$5\overline{Q}$ "	"	"	33	13	"	"	"	"

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TABLE XV. Propagation delay time from set and clear (t_{PLH} and t_{PHL}) (minimum set and clear pulse width check) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T10	$6\overline{Q}$ Propagation delay time from clear (t_{PLH}), C_L = 150 pF	19	30	32	11	8	42	ns	21,23
T11	$7\overline{Q}$ "	"	"	38	18	"	"	"	"
T12	$8\overline{Q}$ "	"	"	36	35	"	"	"	"
U1	1Q Propagation delay time from clear (t_{PHL}), C_L = 50 pF	"	"	5	Open	15	60	"	"
U2	2Q "	"	"	23	21	"	"	"	"
U3	3Q "	"	"	29	8	"	"	"	"
U4	4Q "	"	"	26	Open	"	"	"	"
U5	5Q "	"	"	13	33	"	"	"	"
U6	6Q "	"	"	11	32	"	"	"	"
U7	7Q "	"	"	18	38	"	"	"	"
U8	8Q "	"	"	35	36	"	"	"	"
U9	1Q Propagation delay time from clear (t_{PHL}), C_L = 150 pF	"	"	5	Open	17	68	"	"
U10	2Q "	"	"	23	21	"	"	"	"
U11	3Q "	"	"	29	8	"	"	"	"
U12	4Q "	"	"	26	Open	"	"	"	"
U13	5Q "	"	"	13	33	"	"	"	"
U14	6Q "	"	"	11	32	"	"	"	"
U15	7Q "	"	"	18	38	"	"	"	"
U16	8Q "	"	"	35	36	"	"	"	"

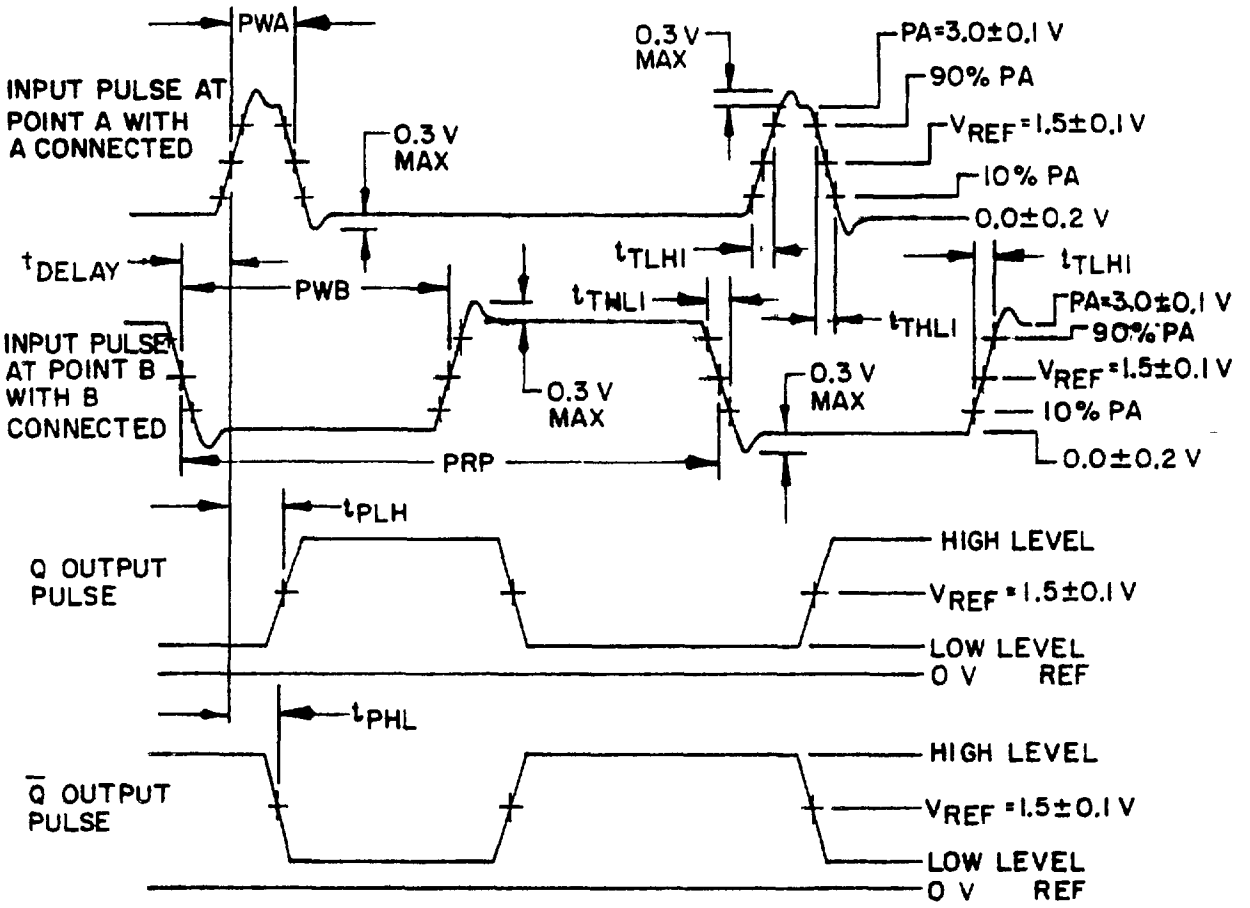
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NOTE: R1 = R2 = 51.1Ω ±1%, 1/4 W.

* FIGURE 21. Propagation delay time from set and clear (t_{PLH} and t_{PHL}) (minimum set and clear pulse width check) test circuit for table XV.

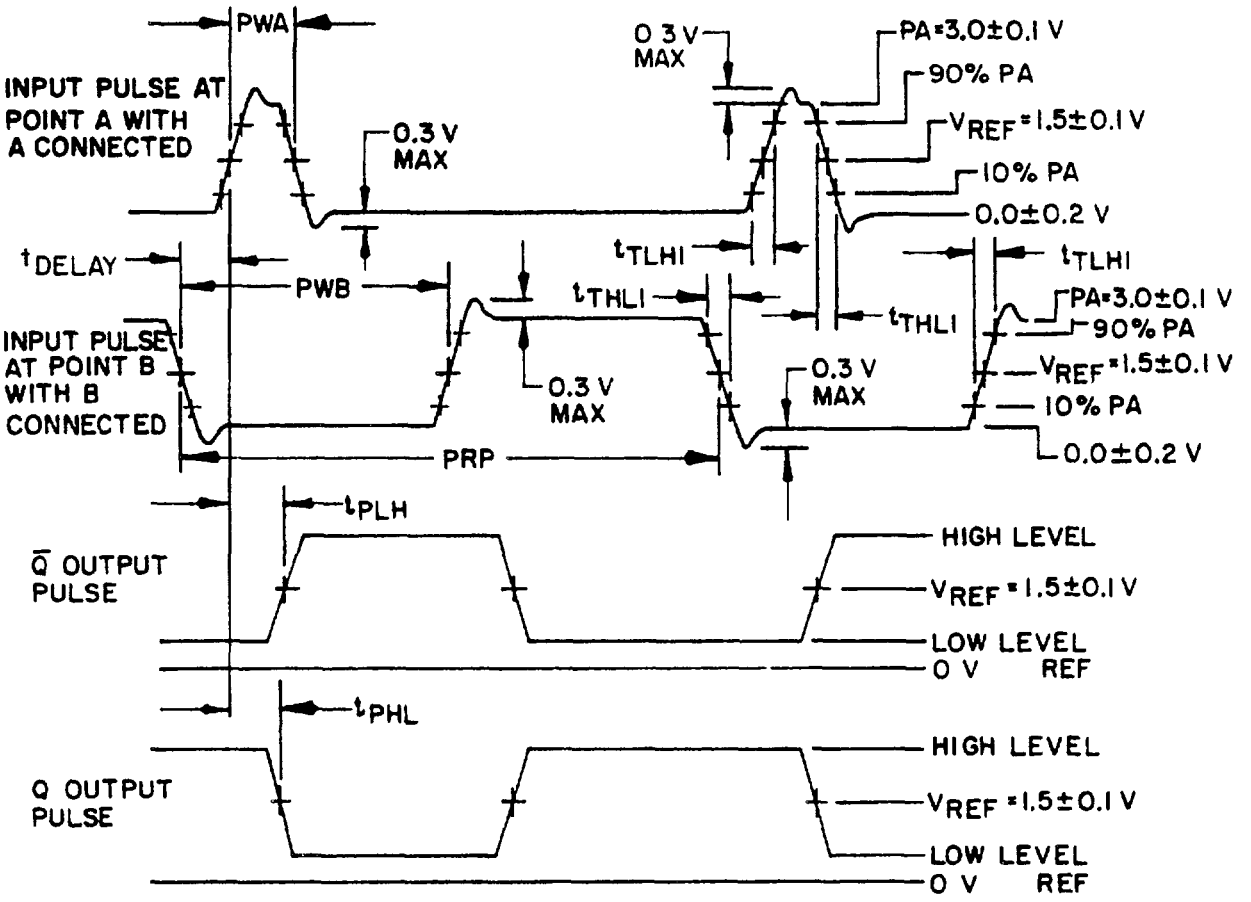
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NOTES:
1. Input pulse parameters (item J01-1):
   PWA = See note 3
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
2. Input pulse parameters (item J01-2):
   PRP = 500 ±50 ns.
   PWB = 250 ±25 ns.
   $t_{DELAY} = 50 \pm 5$ ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
3. Minimum set and clear pulse width check:
   a. Adjust item J01-1 to obtain PWA = 20 +1 ns, -0 ns (minimum set and clear pulse width).
   b. The pulse width requirement must be verified before making measurements per table XV.
4. Verify that all outputs have a high level $\geq$ 2.4 V and a low level $\leq$ 0.4 V before making measurements per table XV.
5. All voltages are referenced to ground.

FIGURE 22. Propagation delay time from set (t_{PLH} and t_{PHL}) (minimum set pulse width check) test circuit waveforms for table XV.

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NOTES:
1. Input pulse parameters (item J01-1):
   PWA = See note 3
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
2. Input pulse parameters (item J01-2):
   PRP = 500 ±50 ns.
   PWB = 250 ±25 ns.
   $t_{DELAY} = 50 \pm 5$ ns.
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
3. Minimum set and clear pulse width check:
   a. Adjust item J01-1 to obtain PWA = 20 +1 ns, -0 ns (minimum set and clear pulse width).
   b. The pulse width requirement must be verified before making measurements per table XV.
4. Verify that all outputs have a high level $\geq$ 2.4 V and a low level $\leq$ 0.4 V before making measurements per table XV.
5. All voltages are referenced to ground.

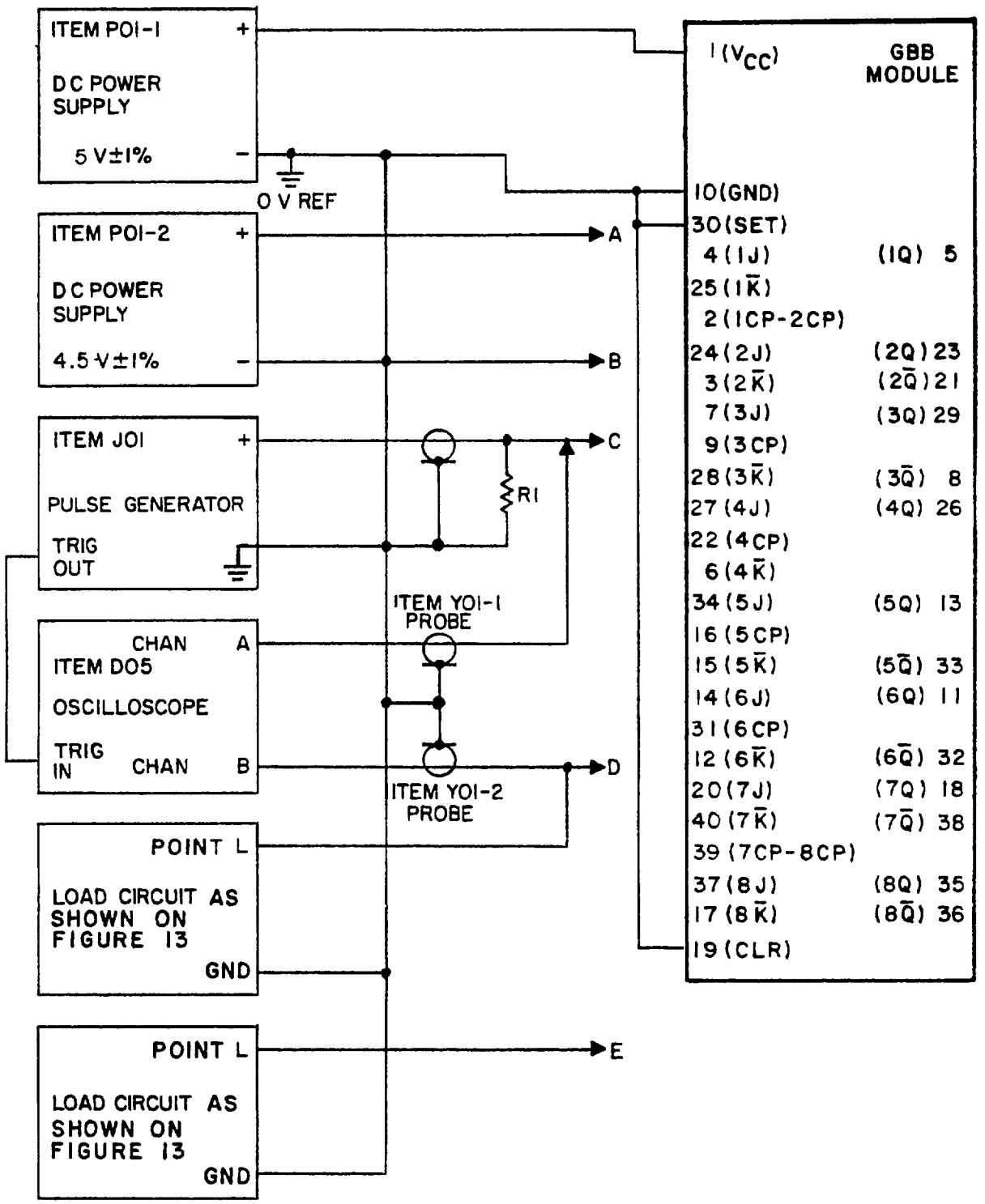
FIGURE 23. Propagation delay time from clear (t_{PLH} and t_{PHL}) (minimum clear pulse width check) test circuit waveforms for table XV.

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TABLE XVI. Maximum operating frequency check (measuring output pulse period - t_{PW}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V1	1Q Maximum operating frequency check (measuring output pulse period - t_{PW}), C_L = 50 pF	4	25	2	5	Open	110	130	ns	24,25
V2	2Q "	24	3	2	23	21	"	"	"	"
V3	3Q "	7	28	9	29	8	"	"	"	"
V4	4Q "	27	6	22	26	Open	"	"	"	"
V5	5Q "	34	15	16	13	33	"	"	"	"
V6	6Q "	14	12	31	11	32	"	"	"	"
V7	7Q "	20	40	39	18	38	"	"	"	"
V8	8Q "	37	17	39	35	36	"	"	"	"
V9	$2\overline{Q}$ "	24	3	2	21	23	"	"	"	"
V10	$3\overline{Q}$ "	7	28	9	8	29	"	"	"	"
V11	$5\overline{Q}$ "	34	15	16	33	13	"	"	"	"
V12	$6\overline{Q}$ "	14	12	31	32	11	"	"	"	"
V13	$7\overline{Q}$ "	20	40	39	38	18	"	"	"	"
V14	$8\overline{Q}$ "	37	17	39	36	35	"	"	"	"

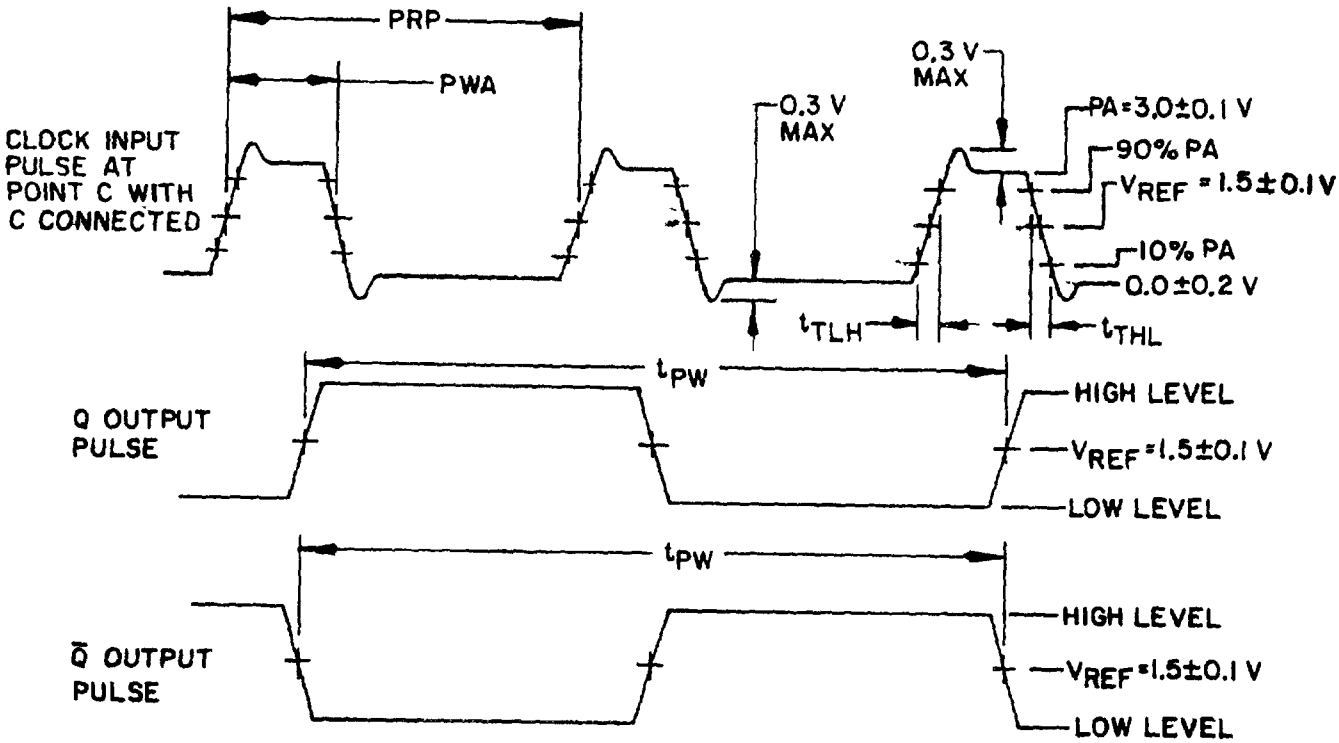
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

* FIGURE 24. Maximum operating frequency check (measuring output pulse period - t_{pW}) test circuit for table XVI.

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NOTES:
1. Clock input pulse parameters:
   PRP = 60 $^{+3\ ns}_{-0\ ns}$ (16.6 MHz $^{+0.0\ MHz}_{-0.8\ MHz}$)
   PWA = 20 $^{+1\ ns}_{-0\ ns}$
   $t_{TLH} = t_{THL}$ = 6 ±1 ns.
2. Verify that all outputs have a high level ≥ 2.4 V and a low level ≤ 0.4 V before making measurements per table XVI.
3. All voltages are referenced to ground.

FIGURE 25. Maximum operating frequency check (measuring output pulse period-t_{PW}) circuit waveforms for table XVI.

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TABLE XVII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W1	1Q Output crosstalk, low level	7,14,20, 24,27,34, 37	3,4,6,12, 15,17,25, 28,40	5	0	400	mV	26,27
W2	2Q "	4,7,14,20, 27,34,37	3,6,12, 15,17,24, 25,28,40	23	"	"	"	"
W3	3Q "	4,14,20, 24,27,34, 37	3,6,7,12, 15,17,25, 28,40	29	"	"	"	"
W4	4Q "	4,7,14,20, 24,34,37	3,6,12, 15,17,25, 27,28,40	26	"	"	"	"
W5	5Q "	4,7,14,20, 24,27,37	3,6,12, 15,17,25, 28,34,40	13	"	"	"	"
W6	6Q "	4,7,20,24, 27,34,37	3,6,12, 14,15,17, 25,28,40	11	"	"	"	"
W7	7Q "	4,7,14,24, 27,34,37	3,6,12, 15,17,20, 25,28,40	18	"	"	"	"
W8	8Q "	4,7,14,20, 24,27,34	3,6,12, 15,17,25, 28,37,40	35	"	"	"	"
W9	$2\overline{Q}$ "	3,4,7,14, 20,24,27, 34,37	6,12,15, 17,25,28, 40	21	"	"	"	"
W10	$3\overline{Q}$ "	4,7,14,20, 24,27,28, 34,37	3,6,12, 15,17,25, 40	8	"	"	"	"

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

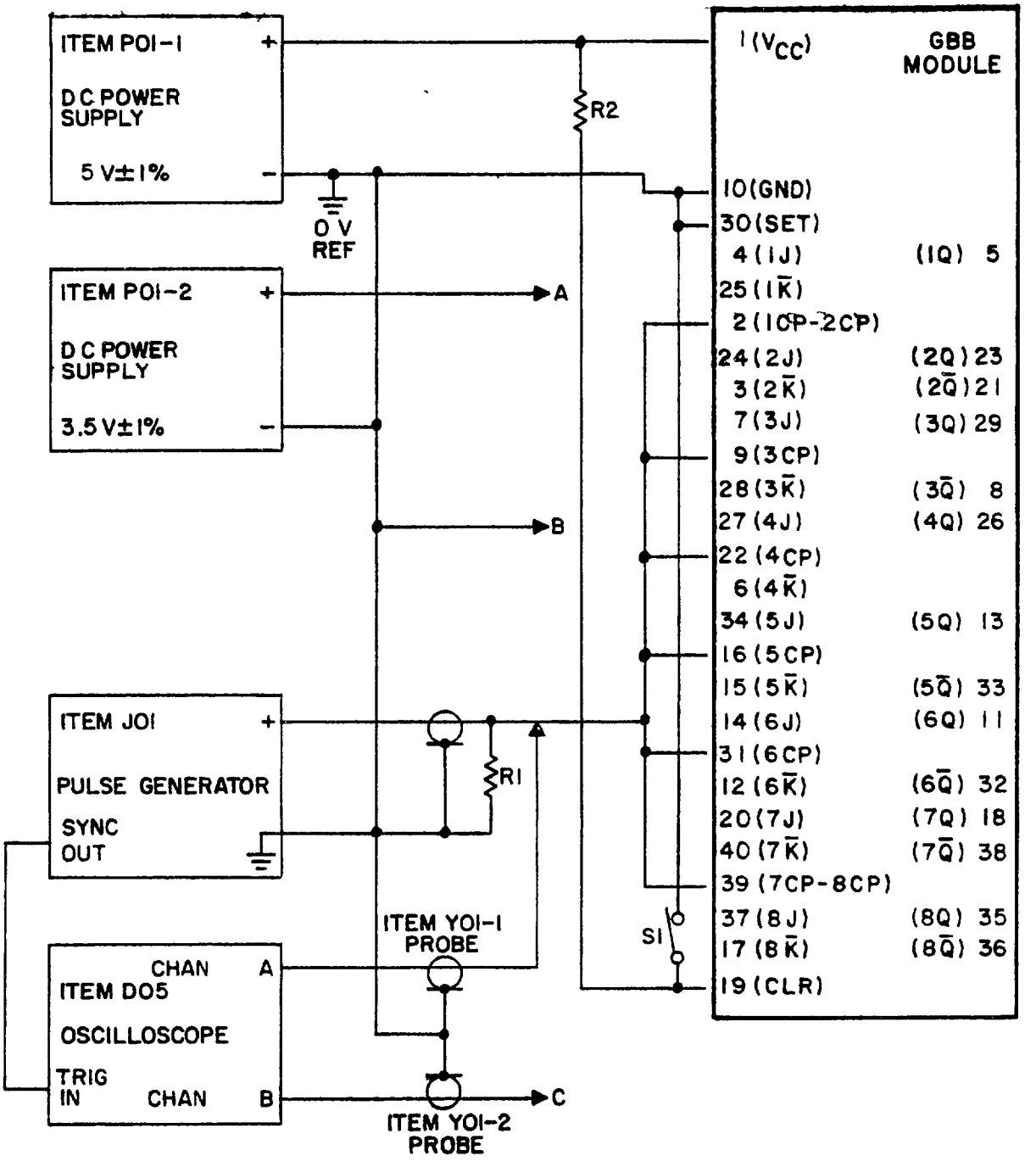
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
W11	$5\overline{Q}$	Output crosstalk, low level	4,7,14,15, 20,24,27, 34,37	3,6,12, 17,25,28, 40	33	0	400	mV	26,27
W12	$6\overline{Q}$	"	4,7,12,14, 20,24,27, 34,37	3,6,15, 17,25,28, 40	32	"	"	"	"
W13	$7\overline{Q}$	"	4,7,14,20, 24,27,34, 37,40	3,6,12, 15,17,25, 28	38	"	"	"	"
W14	$8\overline{Q}$	"	4,7,14,17, 20,24,27, 34,37	3,6,12, 15,25,28, 40	36	"	"	"	"
X1	1Q	Output crosstalk, high level	4,7,14,20, 24,25,27, 34,37	3,6,12, 15,17,28, 40	5	"	"	"	"
X2	2Q	"	3,4,7,14, 20,24,27, 34,37	6,12,15, 17,25,28, 40	23	"	"	"	"
X3	3Q	"	4,7,14,20, 24,27,28, 34,37	3,6,12, 15,17,25, 40	29	"	"	"	"
X4	4Q	"	4,6,7,14, 20,24,27, 34,37	3,12,15, 17,25,28, 40	26	"	"	"	"
X5	5Q	"	4,7,14,15, 20,24,27, 34,37	3,6,12, 17,25,28, 40	13	"	"	"	"
X6	6Q	"	4,7,12,14, 20,24,27, 34,37	3,6,15, 17,25,28, 40	11	"	"	"	"

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TABLE XVII. Output crosstalk (low-level and high-level) test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
X7	7Q	Output crosstalk, high level	4,7,14,20, 24,27,34, 37,40	3,6,12, 15,17,25, 28	18	0	400	mV	26,27
X8	8Q	"	4,7,14,17, 20,24,27, 34,37	3,6,12, 15,25,28, 40	35	"	"	"	"
X9	$2\overline{Q}$	"	4,7,14,20, 27,34,37	3,6,12, 15,17,24, 25,28,40	21	"	"	"	"
X10	$3\overline{Q}$	"	4,14,20, 24,27,34, 37	3,6,7,12, 15,17,25, 28,40	8	"	"	"	"
X11	$5\overline{Q}$	"	4,7,14,20, 24,27,37	3,6,12, 15,17,25, 28,34,40	33	"	"	"	"
X12	$6\overline{Q}$	"	4,7,20,24, 27,34,37	3,6,12, 14,15,17, 25,28,40	32	"	"	"	"
X13	$7\overline{Q}$	"	4,7,14,24, 27,34,37	3,6,12, 15,17,20, 25,28,40	38	"	"	"	"
X14	$8\overline{Q}$	"	4,7,14,20, 24,27,34	3,6,12, 15,17,25, 28,37,40	36	"	"	"	"

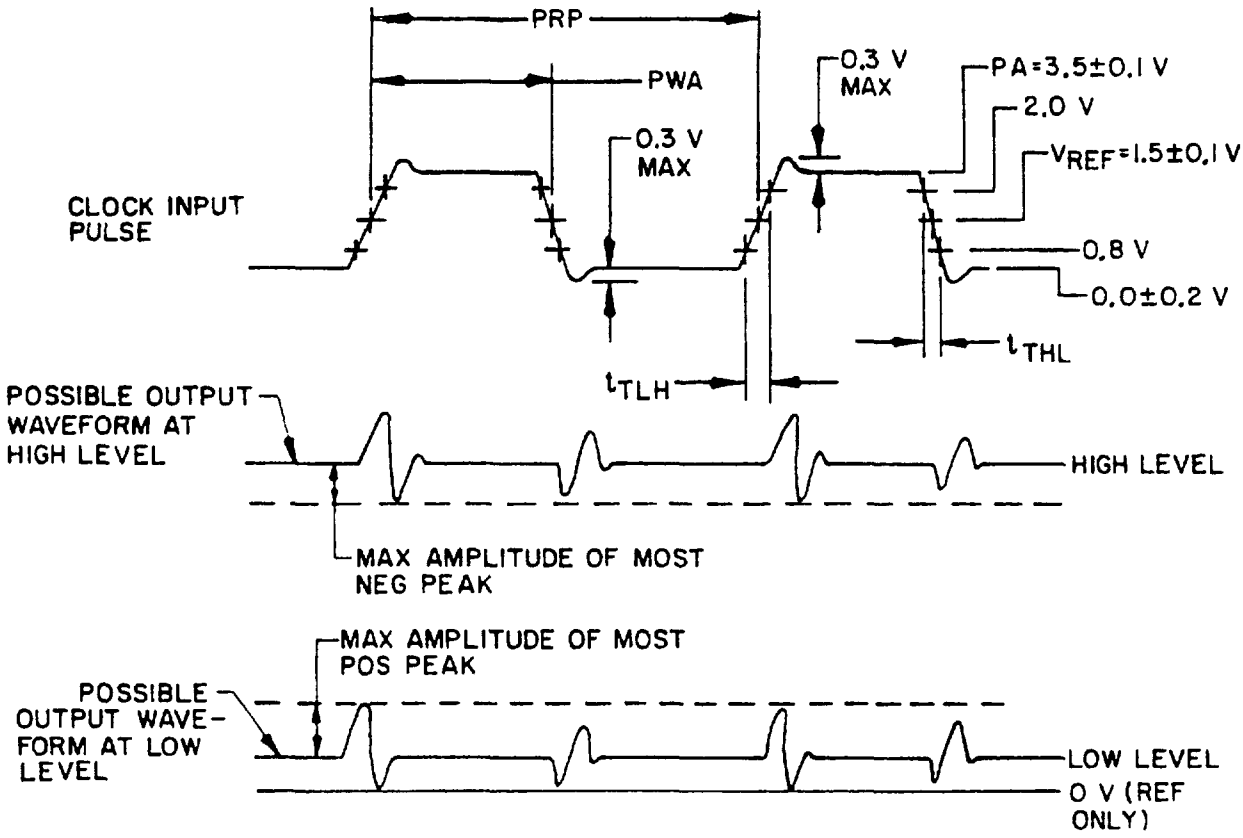
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. R2 = 1 kΩ ±5%, 1/4 W
3. S1 = SPST switch.
4. Load all outputs with load as specified on figure 13 with the exception that capacitor C_L is 50 pF for test table XVII.
5. After making all circuit connections for a test condition open switch S1 for a minimum of 100 ns then close switch S1.

FIGURE 26. Output crosstalk (low-level and high-level) test circuit for table XVII.

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NOTES:
1. Clock input pulse parameters:

   a. PRP = 1,000 ±200 ns
   b. PWA = 500 ±100 ns
   c. $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (measured at the 0.8 V and 2.0 V points of the input pulse).
   d. For each test condition with the module connected to item J01, adjust item J01 for the required pulse amplitude, pulse low level voltage and transition times. Verify the input pulse parameters at each input pin before making each measurement.

2. The lead length of electrical conductors between resistor R1 and input pins, between probes Y01-1 and Y01-2 and input and output pins; between load circuits and output pins; and between ground leads of probes Y01-1 and Y01-2 and test fixture and module ground pins, shall be less than 2 inches (50.8 mm).

FIGURE 27. Output crosstalk (low-level and high-level) test circuit waveforms for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient amplitude	0	400	mV	28,29

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z	Power supply filter capacitance	Open	1	1.6	3.0	µF	30

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TABLE XX. Input capacitance test.

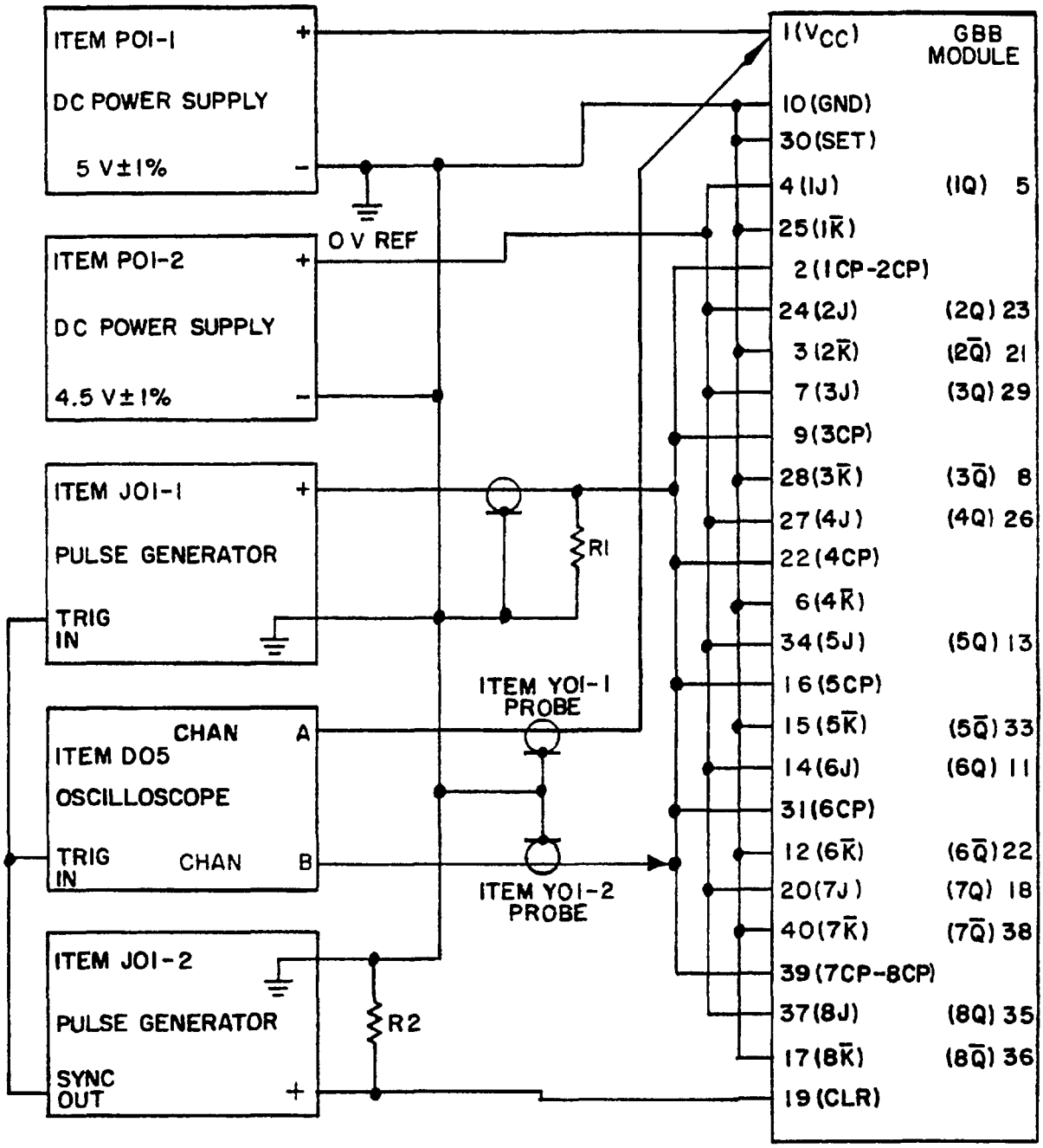
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
AA1	CLR	Input capacitance	19	Open	5	45	pF	30
AA2	SET	"	30	"	"	30	"	"
AA3	1CP-2CP	"	2	"	"	"	"	"
AA4	7CP-8CP	"	39	"	"	"	"	"
AA5	3CP	"	9	"	2	15	"	"
AA6	4CP	"	22	"	"	"	"	"
AA7	5CP	"	16	"	"	"	"	"
AA8	6CP	"	31	"	"	"	"	"
AA9	1J	"	4	"	"	"	"	"
AA10	2J	"	24	"	"	"	"	"
AA11	3J	"	7	"	"	"	"	"
AA12	4J	"	27	"	"	"	"	"
AA13	5J	"	34	"	"	"	"	"
AA14	6J	"	14	"	"	"	"	"
AA15	7J	"	20	"	"	"	"	"
AA16	8J	"	37	"	"	"	"	"
AA17	$1\overline{K}$	"	25	"	"	"	"	"
AA18	$2\overline{K}$	"	3	"	"	"	"	"
AA19	$3\overline{K}$	"	28	"	"	"	"	"
AA20	$4\overline{K}$	"	6	"	"	"	"	"

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TABLE XX. Input capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
AA21	$5\overline{K}$	Input capacitance	15	Open	2	15	pF	30
AA22	$6\overline{K}$	"	12	"	"	"	"	"
AA23	$7\overline{K}$	"	40	"	"	"	"	"
AA24	$8\overline{K}$	"	17	"	"	"	"	"

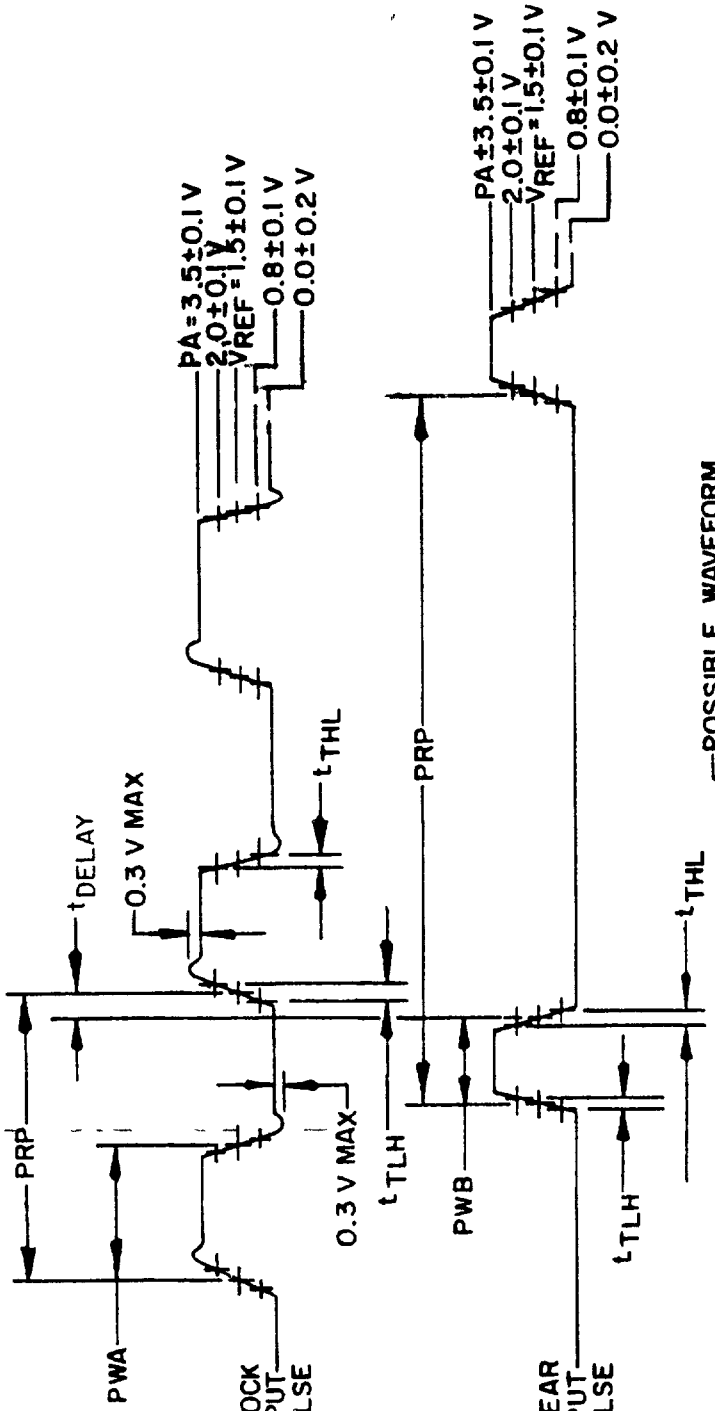
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NOTE. R1 = R2 = 51.1Ω ±1%, 1/4 W.

FIGURE 28. Power supply transient amplitude test circuit for table XVIII.

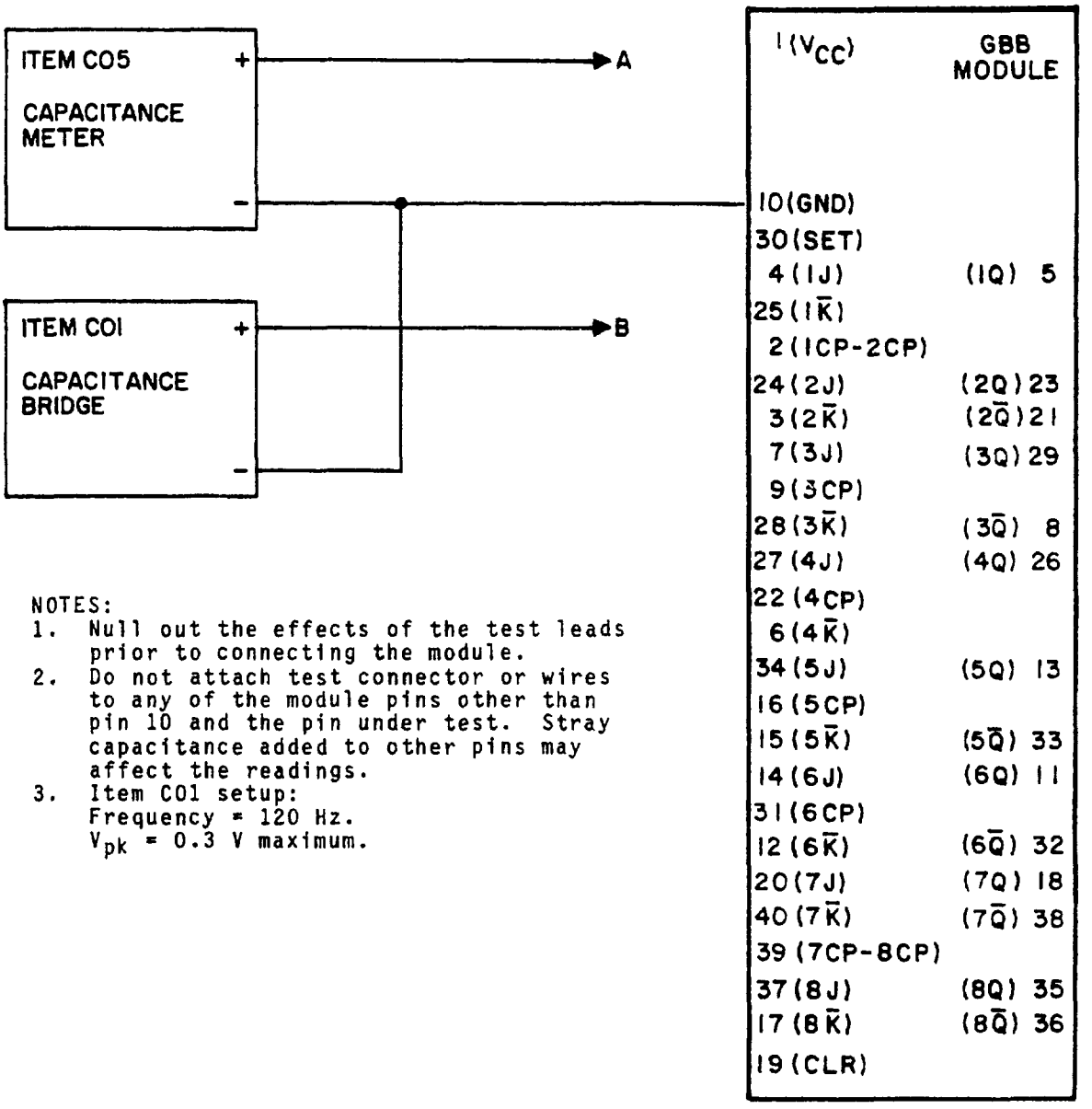
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NOTES:
1. Clock input pulse parameters:
   PRP = 1,000 ±200 ns.
   PWA = 500 ±100 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns.
2. Clear input pulse parameters:
   PRP = 2,000 ±100 ns.
   PWB = 100 ±10 ns.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns,
   t_{DELAY} = 200 ±10 ns.
3. For each test condition with the module connected to items J01-1 and J01-2, adjust items J01-1 and J01-2 for the required pulse amplitude, pulse low level voltage and transition times. Verify the input pulse parameters at each input pin before making each measurement.

FIGURE 29 Power supply transient amplitude test circuit waveforms for table XVIII.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item COI setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 30. Power supply filter and input capacitance test circuit for tables XIX and XX.

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